

**KENYATTA UNIVERSITY
INSTITUTE OF OPEN LEARNING**

AGE 201: PHYSICAL GEOGRAPHY II

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AGE 201: PHYSICAL GEOGRAPHY II

LESSON 1: INTRODUCTION TO PHYSICAL GEOGRAPHY II

1.1 Introduction

In Physical Geography I, we learned that the earth's physical environment is made up of four main subsystems; the lithosphere, the atmosphere, the hydrosphere and the biosphere. The lithosphere and the atmosphere were introduced in Physical Geography I.

In this course, we are going to examine two components of the Earth's physical system; the Hydrosphere and more of the lithosphere.

The **Hydrosphere** is the physical environment occupied by water in all its forms. This includes; precipitation, evaporation, surface runoff, groundwater, rivers, lakes, and oceans with associated processes. This lesson introduces the hydrosphere.

1.2 Objectives

By the end of the lesson you should be able to:

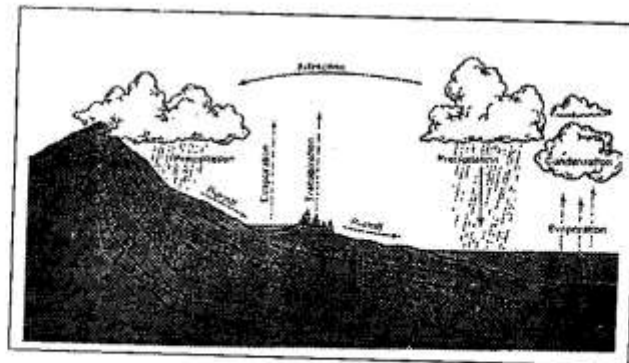
1. Draw a diagram showing the components of the hydrologic cycle.
2. Explain the components of the Hydrologic cycle.
3. Evaluate the water balance concept
4. Give an account of the global distribution of resources

1.3 World Hydrological Cycle

The Hydrological cycle is a continuous process by which water is transported from the oceans to the atmosphere to the land and back to the oceans. Many sub-cycles exist, for example over land, water may evaporate and precipitation may occur before the water returns to the ocean. The complete cycle is global in nature.

Water quantity is finite and indestructible as the global hydrological cycle is a closed system.

Fig 1: The Hydrological Cycle.



The Earth - Atmosphere - Ocean system depicts the earth as the user of water, the atmosphere is the deliver and the ocean is the source.

Water on earth exists in a space called the hydrosphere, which extends on average 15 km upwards into the atmosphere and about 1 km downwards into the lithosphere. Under natural conditions, the hydrologic cycle can be continuous with no beginning nor end.

However, it can be perceived to start with the evaporation process in which water is transformed from the liquid phase on land and in the oceans to the gaseous phase becoming water vapour in the atmosphere. Through the process of condensation, water vapour in the atmosphere is converted into the liquid phase from which it may fall as precipitation. On reaching the ocean, the precipitation may re-evaporate again. Precipitation on the land interacts with the surface in many ways taking different paths within the water cycle. It may be intercepted by vegetation surface, or on reaching the ground, it may infiltrate through the surface layer into the soil, or it may flow over the ground as overland flow.

Plants through the process of evapo-transpiration may evaporate water in the soil layer, while that which is not evaporated may stay as soil moisture in the soil or flow laterally as interflow re-surfacing into surface flow or percolating to the aquifers. Overland water reaches the channels and flows to the sea through gravity.

The movement of water in its various forms requires energy and the main driving forces of the hydrologic cycle are; radiation and gravity. Radiation from the sun provides the energy for evapo-transpiration and generates air movements, which can move moist air over the oceans to the land. Gravity in turn, acts on precipitation and water flow, driving water down slope toward the oceans or seas. Movement continues at all altitudes, in lakes, and oceans under forces created by temperature differences and Earth's rotation (Coriolis force).

1.4 'The Water balance Concept

The movement of water in any part of the hydrologic cycle may be quantified in terms of water balance equation. This equation considers change in storage as a difference between inputs and outputs.

$$\text{Incoming Water Volume} = \text{Outgoing Water Volume} + \text{Change in Storage}$$

$$I = O \pm \Delta S$$

Where

I = Inflows e.g., precipitation, discharge into an hydrologic system.
(Lake, wetland, pond, dam etc).

O = Outflows e.g., evapotranspiration, discharge from the system.
(Lake, wetland, pond, dam etc).

ΔS = Changes in Storage

We note that the equation is applicable over a period of time, usually an hydrological year.

1.5 Distribution Of Global Water

The total volume of water in the global water cycle is constant, but the distribution and movement through the cycle are erratic both in time and space. At times, the movement of water is extremely rapid as in torrential rains and floods; at other times there is no apparent movement, especially during drought or within ice caps.

The annual precipitation on land constitute about 0.008 % ($119,000\text{km}^3$) while the annual evapo-transpiration from the land to is 0.005 % ($72,000\text{km}^3$) of the total global water volume.

The annual surface runoff into the oceans constitutes $47,000\text{km}^3$ that is 0.003 %. Table 1 gives estimated quantities of water in its different forms and their estimated residence times in the hydrological cycle.

	Area (10^6km^2)	Volume (km^3)	Percentage of total	Residence times
Oceans	361.3	1,338,000,000	96.5	2000-4000yrs
Ground water	269.6	23,400,000	1.69	300yrs
Soil moisture	82	16,500	0.0012	2 weeks -1yr
Polar ice	16.0	24,023,500	1.7	~1000yrs
Lakes	2.0	176,400	0.013	~10
Rivers	148.8	2,120	0.0002	~2 weeks
Atmospheric water	510.0	12,900	0.001	10 days
Total water	510.0	1,385,984,610	100	

1.6 Summary

The hydrosphere entails all the areas covered by water. Water movement in the hydrosphere is summarized by the hydrologic cycle. The cycle shows how water moves from the earth's surface, into the atmosphere and back to the earth's surface. Solar energy and force of gravity drives the hydrologic cycle. The water balance concept illustrates that for any hydrologic system, change in storage is effected by the inflow and outflow over a specified time period. Global water exists in different forms and proportions. This includes, oceans, ground water, soil moisture, polar ice, lakes, rivers and water vapour.

1.7 Revision Questions

1. Using suitable illustration, describe the components of the hydrologic cycle.
2. Explain the water balance concept using a fishpond or a dam.
3. What would you comment about the global distribution of hydrologic cycle components?

1.8 Further Reading

1. Shaw, E. M. (1998). **Hydrology in Practice**. Prentice Hall, London.
2. McNight, H. (1995). **Physical Geography, A Landscape Appreciation**. Prentice Hall, Los Angeles.
3. Doherty, A., and McDonalty, M. (1999). **River Basin Management**. Hodder & Stoughton, London.
4. Strahler, A.H. and Oberlander, T.M. (1992). **Modern Physical Geography**, John Wiley and sons.
5. Chorley R. J. & Barry R. G. (1976), **Atmosphere, Weather and Climate**, Methuen Co. Ltd, London.

LESSON 2 FORMATION OF PRECIPITATION

2.1 Introduction

In lesson 1, elements of the Hydrologic Cycle and the global water balance were examined. In this lesson, precipitation as a component of the hydrological cycle is explained. Its formation and characteristics are discussed.

2.1 Objectives

By the end of this lesson, you should be able to:

1. Define the various forms of precipitation
2. Explain the mechanisms of precipitation formation
3. Account for the spatial and temporal variability of precipitation.

2.3 Evaporation

Before precipitation is formed, evaporation has to take place. Heating of water, which then is transformed from liquid to gaseous state, causes the process of evaporation, driven by solar energy. Water vapour comprises approximately 1-4% of the atmosphere and remains in the atmosphere for about 10 days (see table 1). The global annual average evaporation rate from the oceans is estimated to be 1400mm (505,000 km³/yr), while the transpiration rates are 484mm (72,0000 km³/yr).

Since air can only hold a finite amount of water vapour at any given temperature, it is said to be saturated when maximum water vapour has been absorbed. Note that the higher the temperature, the higher the amount of water it can hold. Condensation occurs and clouds are formed when maximum limit is reached either by decrease of temperature or increase of vapour pressure.

Atmospheric circulation caused by global temperature differences moves moisture-laden air and clouds around the planet. The process of evaporation and condensation are crucial in the air conditioning of the earth. Evaporation uses up energy and cools the atmosphere, while condensation releases energy and warms up the atmosphere. Evaporation also

purifies seawater as dissolved solids and organic and inorganic constituents are left behind when the water vapour is purified.

2.4 Condensation and Cloud Formation

Clouds are made up of millions of very small drops of liquid water or ice crystals that are too light to fall out of the atmosphere. The cloud particles form from air that contains water vapour when the temperature falls to a critical level called the dew point. These liquid droplets may then freeze into ice crystals. Before water or ice crystals can form, two things must happen;

- Moist air must rise, reducing its pressure and giving up its heat to the surrounding atmosphere.
- Dust particles must be present on which the cooled vapour can condense to form droplets or ice crystals. These tiny particles are called condensation nuclei and Ice nuclei.

Two other mechanisms that account for precipitation-sized particles are; the Bergeron-Findeisen process or ice-crystal process, and the Coalescence process. In the coalescence process, we have collision and merging of water droplets falling at different velocities. This happens when large droplets falling faster than the small ones, take them along unified. In convection clouds over tropical oceans, rain is produced entirely by coalescence process. Clouds have bases at an altitude of about 600m and cloud tops can reach up to 3 km.

In the ice crystal process, ice crystals and super-cooled water droplets co-exist and droplets grow while droplets evaporate.

In clouds that contain both ice crystals and droplets or super-cooled water (water at temperatures less than 0°C), the droplets evaporate and the vapour condenses to form ice crystals. Thus, the crystals grow at the expense of the droplets until they are large enough to fall out of the cloud. If they melt on the way down as it often happens, rainfall is observed at the ground. If the cloud contains no ice crystals, precipitation particles grow

by the coalescence of different sized droplets as they fall through the cloud. The larger a drop becomes, the more efficient it is at collecting smaller ones and the greater its chance of reaching the ground.

The two basic shapes of clouds are in layers or in heaps. These are as a result of two different ways in which air can move upwards. When air rises slowly over large areas at rates of a few centimeters per second, layers of stratified clouds are formed. In the mid-latitude regions, this frequently occurs in cyclones, especially in warm sectors and at warm fronts. Rapidly rising air (several metres per second) occurs in convectional currents, which are usually only a few hundred meters across and near the ground. These currents widen with altitude but the resultant heaped or cumuliform clouds are rarely more than a few km across. If the atmosphere is unstable, they may grow into very large cumulonimbus clouds.

A cloud is identified by its shape and height above the ground. The highest clouds (8-10km) are made up of ice and are called cirrus, cirrostratus and cirrocumulus; middle clouds (3-8km) are made up water and ice, and are called alto cumulus and altostratus; and the low clouds below 3km are made of water and are called stratus, strato-cumulus and nimbostratus. Two other types are cumulus and cumuli-nimbus. It is important to note that clouds of different types often occur together.

2.5 Precipitation Types and Characteristics

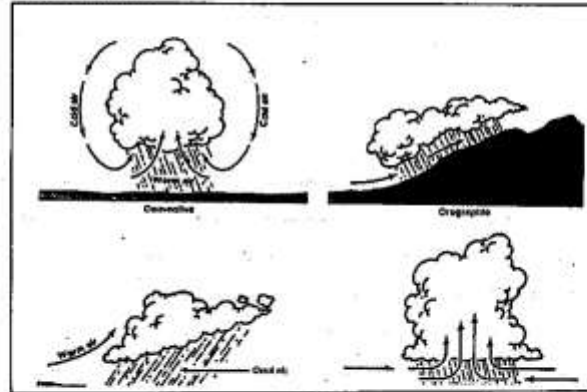
Various forms of precipitation exist, namely rain, snow, hail, fog, drizzle, etc. Rain refers to all liquid forms of precipitation ranging in diameter from 0.5mm to 5mm.

Mechanisms triggering condensation and potential for precipitation include;

- Convergence of contrasting air masses
- Local heating and rising of air masses (thermals)
- Presence of physical barriers such as mountains.

Precipitation from clouds occurs when the condensed water droplets become large enough to overcome gravity. The source mechanism gives rise to rainfall of different characteristics. These range from high intensity convective storms of short duration caused by local heating of air masses common in Tropics, to low intensity drizzles of long duration, associated with convergence of a warm air mass over a cold air mass, typical of the temperate mid-latitudes. Figure 2 show the convective and cyclonic rain.

Figure 2: Types of Precipitation



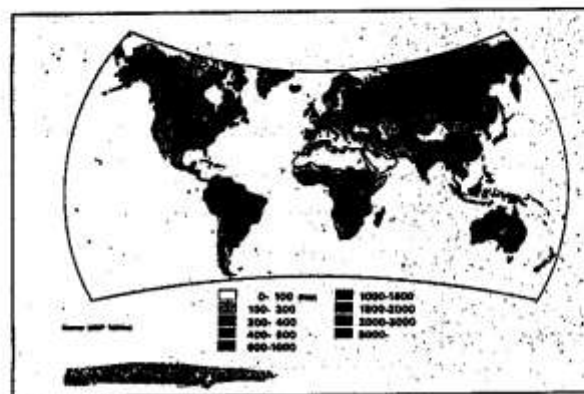
Water can also precipitate through the process of condensation directly on vegetative and earth surfaces especially due to overnight cooling above the ground yielding to dew. The mechanisms for rainfall formation include; convection, orographic lifting, frontal lifting, tropical depressions or cyclones and convergence. In convection, the air is said to be unstable due to the vertical instability of the air having temperature gradient greater than the adiabatic lapse rate. A parcel moving up obtains a temperature higher than its surroundings. Convective rainfall is common in tropical regions and appears as thunderstorms in temperate climates during the summer period. Rainfall intensities of convective storms can be very high locally, but the duration is short.

In **orographic lifting**, air passing over a mountain may be caused to rise and rainfall caused in the windward slope with highest amounts in the mountaneous part of the basin. In frontal lifting, the existence of an area with low pressure causes surrounding air to move into the depression, displacing low pressure air upwards which may then be cooled to dew point. If cold air is replaced by warm air, the frontal zone is usually large and rainfall of low intensity and long duration. A cold front would usually have steeper slope..

For convergence, the Inter Tropical Zone is the region where the air masses originating from the tropics of Cancer and Capricorn converge and lift. The presence of the ITCZ governs the occurence of wet and dry seasons.

The global variation of precipitation distribution on annual basis is depicted in the Figure 3.

Figure 3: Mean Annual Global Precipitation



Evidently, most of the rainfall occurs on Equatorial and Monsoon zones. It is lowest in the desert regions of the tropics and the poles.

Orographic precipitation is due to mountains exposed to prevalling winds. Leeward slopes, being on the rain shadow receive less rainfall. However, it is noteworthy that precipitation only increases with altitude up to some height, above which rainfall

decreases as air masses lose most of their moisture at lower altitudes. In addition to spatial distribution of the annual precipitation, temporal variations occur with distinct seasonal patterns characteristic of some climatic zones. This is important for the hydrologic cycle and man's utilization of water resources.

2.6 Summary

In this lecture, we have looked at the process of precipitation formation, and its position in the Hydrologic Cycle. We have seen the various factors that influence precipitation formation. We have also established the various forms of precipitation.

2.7 Revision Question and Activities

1. Why is the study of precipitation important?
2. (a) What is the significance of condensation nuclei in precipitation formation?
(b) Explain mechanisms of precipitation formation.
3. Activity
 - Observe what happens in your area during a rainy day. Enumerate the stages of the formation and sources of moisture.
 - Attempt the variation in time during the storm and the monthly variations for a station in your district.
 - How do these variations influence the socio-economic activities in your area?

2.8 Definition of key terms

Precipitation: This is the deposition of moisture on the land's surface generating from the atmosphere in the form of rain, sleet, snow, dew or hail.
Condensation: This is the physical process through which vapour passes into the liquid or solid form that occurs as the air is cooled to its dew point or air becomes saturated through evaporation into it.
Rainfall intensity: This is the amount of rainfall that occurs within a given time interval usually stated as mm/hour.
Dew point: This is the critical temperature in which when cooled air can become saturated with water vapour and is usually measured using a hygrometer.

2.7 Further Reading

1. Shaw, E. M. (1998). *Hydrology in Practice*. Prentice Hall, London.
2. McNight, H. (1995). *Physical Geography, A Landscape Appreciation*. Prentice Hall, Los Angeles.
3. Doherty, A., and McDonald, M. (1999). *River Basin Management*. Hodder & Stoughton, London.
4. Knap B. J. (1981). *Practical Foundations of Physical Geography*. George Allen and Unwin, London.
5. Niewolt, S (1977). *Tropical Climatology, An Introduction to the Climate of the low Latitudes*. John Wiley & sons, Chichester.
6. Gregory K, J (1985). *The Nature of Physical Geography*. Edward Arnold, London.

LECTURE 3: USE OF WATER BY VEGETATION

3.1 Introduction

This lesson explains the role of vegetation in the hydrologic cycle. Vegetation influences water loss into the atmosphere through evapotranspiration. Plants encourage the rate of water movement into the soil, i.e. infiltration. Vegetation water use is also influenced by moisture availability in the soil.

3.2 Objectives

By the end of the lesson, you should be able to:

1. Elucidate the role of vegetation in the hydrologic cycle.
2. Explain how vegetation cover influences infiltration.
3. Explain the significance of water to vegetation.

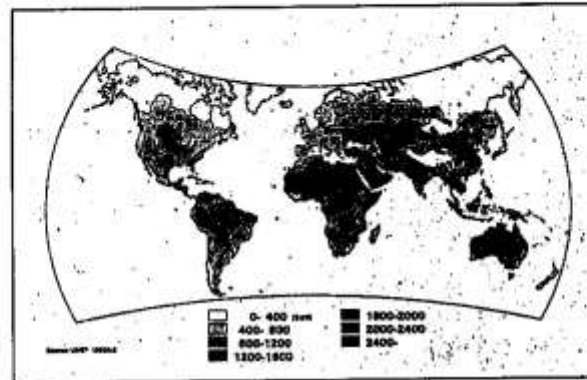
3.3 Hydrological Significance Of Vegetation

The hydrological significance of vegetation can be studied by investigating the evapo-transpiration and infiltration processes in an area. Vegetation and soils play an important role in determining the different pathways rainfall may take when it reaches the ground. Evaporation and transpiration (evapo-transpiration) are a function of vegetation conditions among other factors such climatic conditions and soil water.

When water is readily available, the evapo-transpiration rate is driven by the needs of the plant and climatic factors. The climatic factors are; solar energy, temperature, atmospheric humidity and wind speed. This, we call potential evapo-transpiration (PET) rate.

Two major factors that affect PET rate are latitude and altitude. In the Tropics, PET rates at sea level reach 7-8mm/day (2500mm/yr) while at an altitude of 2000 meters, the same rate drops to 3-5 mm/yr (1100mm/yr). At higher latitudes summer PET rates are approximately 4-5mm/day at sea level, dropping to almost zero in winter. This results into an annual average rate of 600-800mm/yr. This is illustrated in Figure 4.

Figure 4: Mean annual Global potential evapo-transpiration (mm)

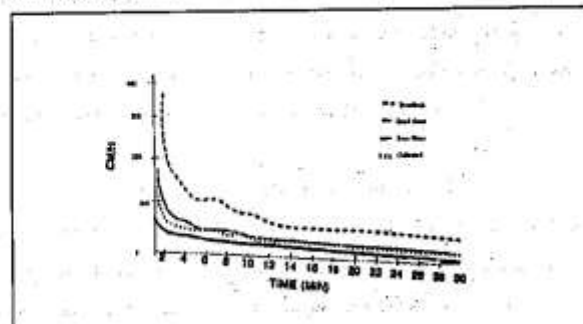


When the water supply is limited, the rate of evapo-transpiration drops below the potential rate to Actual Evapo-transpiration (AET). The AET is a function of availability of soil water, plant species, vegetation cover and plant growth. In general, the high rainfall and evapo-transpiration rates of the tropics give rise to faster vegetation growth and biomass production than found in the higher latitudes.

3.4 Effect of Vegetation on Infiltration Process

Infiltration rate is the process by which water moves into the soil through the surface layers. Vegetation cover and soil properties will determine the rate of infiltration and amount of water that can be stored in the soil. The presence of plant roots promotes infiltration. Generally, higher infiltration rates are associated with surfaces vegetation cover than surfaces with bare soil as depicted in Figure 5.

Figure 5: Infiltration Rates for Different Soil and Vegetation covers.

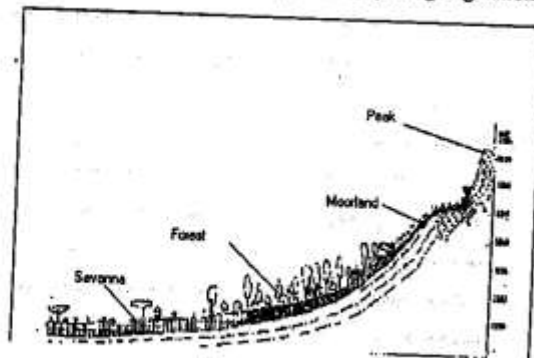


The protective capacity of vegetation is particularly important in the tropics where intense and erosive storms are common. The rate of infiltration depends on both soil properties, vegetation of the surface layer as well as the intensity and distribution of the incoming rainfall and on the sub-surface soil drainage conditions.

Within the rooting depth of plants, the soil moisture is the source of water for existing vegetation between rainfall events. Since the moisture is held under tension, plants have to exert a force to extract the water through their roots for transpiration. The solar energy acting through the process of photosynthesis provides this force. Rate of photosynthesis is a function of temperature. The growth of plants can be constrained when water is not easily removable and plants are forced to reduce their transpiration and growth rates.

Desert plants can exert much higher suction force than wetland plants that are sensitive to water shortages. Soils can store up to 250-300mm of available water. Saturated or waterlogged soils are susceptible and can only be tolerated by certain plants and vegetation types especially swamps, wetlands and floodplains. Although only 0.003% of water is stored in plant tissues, vegetation has high turnover of water through transpiration. The average residence time of water in plants is only a few days. For soils residence times is from a few weeks to a year. A variety of plants communities may develop in response to local environmental conditions. The spatial distribution of plants/vegetation types is as a result of special adaptation to conditions such as altitude, climatic controls, soil conditions and local topography. The cross section of the Mt. Kenya transect (Figure 6) depicts the different vegetation/ land cover types at different levels.

Figure 6: Transact of Mt Kenya area depicting vegetation/cover at different altitudes



3.5 Significance of Water to Vegetation

The study is vital to those countries where irrigation is used to improve food yields. Soil water is a critical medium for supplying nutrients to growing plants. Plant nutrients such as calcium, potassium, nitrogen and phosphorus are absorbed in solution.

Water availability in vegetation influences the opening and closing of the stomata. When soil water is limited, the stomata will close to reduce water loss. Some plants will lose their leaves as a means of reducing water loss.

3.6 Summary

Through the process of transpiration, soil water is transferred into the atmosphere in the form of vapour by vegetation. When precipitation falls, vegetation intercepts part of it and this may evaporate back to the atmosphere. The rate of water movement into the soil is high in areas with vegetation though it varies from one plant species to another. Soil water influences a number of vegetation processes such as nutrient movement and the opening and closing of the stomata.

3.7 Revision Questions

1. Explain the effect of vegetation on infiltration rates
2. Explain how clearing of vegetation has interfered with the hydrologic cycle
3. What is the role of infiltration in agriculture? Indicate the methods employed to improve infiltration in farms.

3.8 Key terms

Infiltration – The downward entry of water into the soil surface.

Infiltration rate – The maximum rate at which water can enter the soil under specified conditions. An infiltration curve can be plotted to show the rate of infiltration.

3.9 Further Reading

1. Bergsma, E. (1996). **Terminology for Soil Erosion and Conservation**. FAO, Italy.
2. Liniger, H. (1995). **Endangered Water**, GDE., Berne-Switzerland.
3. Shaw, E. M. (1998). **Hydrology in Practice**. Chapman & Hall, London.
4. Graham Waters (1979) **First Lessons in Physical Geography**, Longmans, London.

LECTURE 4: GROUND WATER

4.1 Introduction.

This topic explores underground water that forms part of the hydrologic cycle. Various aspects of ground water such as recharge, discharge, water table and aquifers have been explained. The significance of ground water is also highlighted.

4.2 Objectives

- | |
|---|
| By the end of the topic you should be able to: |
| (i) Define ground water resources |
| (ii) Explain the conditions for ground water occurrence |
| (iii) Discuss the significance of underground water. |

4.3 Definition of Ground water

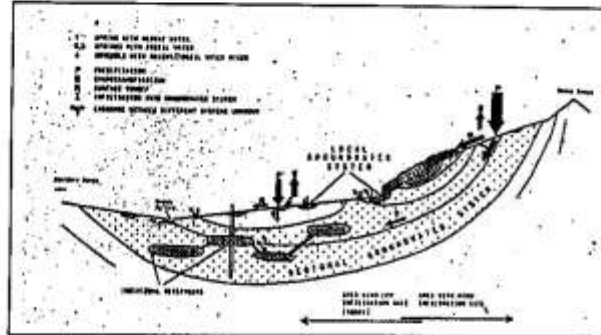
Ground water may be defined as sub-surface water in soil and rocks that are fully saturated. Under force of gravity, sub-surface water not used by plants or held by tension in the unsaturated zone, and which is not subject to lateral flow will percolate vertically filling up the porous volume of the geologic layers forming ground water.

The rock formations bearing ground water are called aquifers. Ground water may be defined as sub-surface water in soil and rocks that are fully saturated. Pore spaces are filled with water and the zone is saturated. The boundary between the unsaturated and saturated zones is known as the water table. It is also known as the phreatic surface. At the water table, pressure is at atmospheric level. Aquifers that have such a water table are known as unconfined aquifers.

Ground water is subject to gravity and thus flows from areas where the water table is higher to areas where it is lower. Where ground water flows intersect with the land surface, it contributes to stream-flow.

Ground water is recharged from rainfall over areas with permeable soils and rocks. Groundwater flow may be directed to strata below impermeable rocks where it is confined under pressure. Here we have, confined aquifers. Figure 7 shows recharge and other ground water concepts

Figure 7: Ground water recharge and other concepts.



4.4 Aquifers

Aquifers are geological formations comprising layers of rock or unconsolidated deposits that contains sufficient saturated material to yield significant water quantities. Most major aquifers are composed of sedimentary deposits formed from erosion and deposition of other rocks. Igneous and metamorphic rocks have few interconnected pore spaces and have very low water bearing capacities.

Aquifers can be confined or unconfined. The upper boundary of the zone of saturation varies differently in the two aquifers. For unconfined aquifers, this boundary is the water table defined as the level where pore water pressure is equal to atmospheric pressure. In the case of confined ground water, the upper boundary of the water body is formed by an overlying less permeable bed.

Most confined aquifers have an area through which recharge to the ground waters occurs by means of infiltration and percolation, and in which a water table as defined above

represents the zone of saturation. Further, continuing impermeable bed rarely forms an absolute barrier to ground water movements. Since the water table in the unconfined ground water area through which recharge takes place is situated at a higher elevation than the confined area of aquifer, it is apparent that the ground water in the latter area is under pressure equivalent to the difference in hydrostatic level between the two.

The amount of ground water stored in a saturated material depends on porosity. This is the percentage of the total volume of a rock that is represented by its interstices or voids. In analyzing aquifer systems, it is common to assume that the aquifer is homogenous and isotropic, i.e., that certain properties such as porosity have the same value in different parts of aquifer (homogeneity) and in different directions from the same point (isotropy). However, apparently uniform deposits may have preferred orientation of particles or fractures (anisotropy).

4.5 Recharge

All ground water is virtually derived from precipitation. The main components of ground water recharge are infiltration of part of the total precipitation at the ground surface, seepage through the bed and banks of surface water bodies such as lakes, rivers and oceans, ground water leakage and inflow from adjacent aquitards and aquifers and artificial recharge from irrigation, spreading operations and leakage from water supply pipelines.

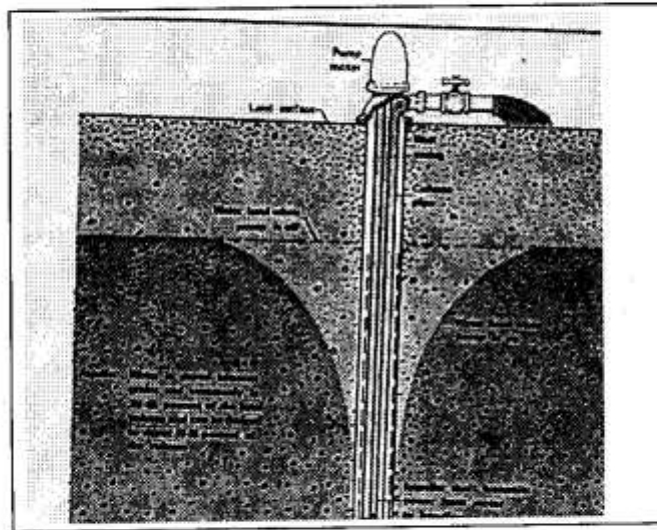
4.6 Discharge

The main components of ground water discharge are evaporation, particularly in low-lying areas where water table is close to the ground surface, natural discharge by means of spring flow and effluent seepage into surface water bodies, ground water leakage and outflow through aquitards and into adjacent aquifers; and artificial abstraction through wells and borehole

The water table depth will determine the rate of evaporation losses. Evaporation will also reduce soil water content and the shallow water table is bound to have diurnal

fluctuations. Effluent seepage is important for sustaining the base flow in rivers during dry weather periods. Ground water is abstracted for many uses, domestic, agricultural and industrial. A typical well installed with a pumping system as illustrated in figure 8.

Figure 8: Typical well abstracting from an aquifer.



Pumping water to the land is costly and the cost depends on the depth of the water, the ease with which water flows through the aquifer and the cost of power. There can be seasonal fluctuations in the water table reflecting changes in aquifer storage. With continuing excess of discharge over recharge, subsidence of the overlying ground surface may result due to compression of the aquifer. The amount of subsidence at any location depends upon the sub-surface lithology, thickness of the compressible units and their storage characteristics and; upon the magnitude and duration of the decline in head.

A good example of the largest recorded subsidence in a geothermal field was at Wairakei in New Zealand where the ground subsided thirteen meters at a maximum subsidence rate of 45 cm/year in a small region outside the production area. In the early stages of a geothermal development, geothermal fluids are withdrawn from a reservoir at a rate

greater than inflow into the reservoir. This net outflow causes rock formations at the site to compact. Particularly in the case of clays and sediments leading to ground subsidence at the surface. The key factors for subsidence are:

- a pressure drop in the reservoir as a result of fluid withdrawal
- presence of highly compressible rock formation above or in part of the reservoir
- presence of high permeability paths between the reservoir and ground surface.

The effect of the Wairakei subsidence included; creation of a pond in what was originally a stream, cracking of nearby highways and compressible buckling and tensile fracturing of steam pipelines and fissures in the surrounding fields. A recent experience of the Baringo geyser for a bore hole attempt for water supply shot above 50 meters high releasing hot water from the ground aquifers.

Deposits of young, poorly consolidated material of high porosity underlie areas prone to subsidence due to ground water extraction, and much of the subsidence may occur due to compression of the clayey aquitards. Ground water movements and factors affecting the flow will be the subject of a third level module.

4.7 Role of Ground Water

Ground water has a big role in the regulation of the hydrological cycle. It has large residence times despite considerable site-to-site variations. Ground water sustains stream-flow during period of dry weather and is a major source of water in arid and semi-arid areas. Due to its long residence time of the order hundreds of years, arid climate areas with little opportunity for percolation to depths may nevertheless have significant ground water reserves which are the result of recharge in the past more pluvial times. There are large ground water reserves in the Sahara. Most of the ground water has its source from precipitation of atmospheric moisture that infiltrates and percolates the zones of aeration.

4.8 Definition of Key Terms

Local ground water is water that percolates beyond the rooting depth of vegetation and recharges the ground water table that feeds local springs and streams. This water remains for short periods before reappearing on the surface to feed rivers, lakes or the sea or other ground water systems.

Regional ground water recharges a ground table which extends beyond the local area. Ground water emerges at some distance from the source of recharge and this water can take years to contribute to surface water systems. It is not easy to distinguish between local and regional ground water systems.

Note During infiltration and percolation processes, the water quality may change due to chemical properties of the soils and rocks through which it passes.

4.9 Revision Questions

1. Define the terms:
 - a. Water table
 - b. Confined and unconfined aquifers
2. Explain the processes of ground water recharge and discharge.
3. Highlight the dangers of unlimited ground water extraction.
4. Explain the significance of underground water resources.

4.10 Further Reading

- 1 Shaw, E. M. (1998). **Hydrology in Practice**. Prentice Hall, London.
- 2 McNight, H. (1995). **Physical Geography, A Landscape Appreciation**. Prentice Hall, Loss Angeles.
- 3 Doherty, A., and McDonalaly, M. (1999). **River Basin Management**. Hodder & Stoughton, London.
- 4 Todd D. K (1982) Ground water Hydrology, John Wiley and sons.
- 5 Trewatha G. T. (1968). Fundamentals of Physical Geography; McGraw Hill.

LECTURE 5 RIVER BASIN HYDROLOGICAL PROCESSES

5.1 Introduction

A river drainage basin, also called a watershed or catchment is the area encompassed by the topographic divide that contributes overland flow and ground water to a specific point in a river. Drainage processes within the basin include; stream flow, overland flow and flooding among other processes. Water balance for a basin or catchment as hydrologic system will be explained.

5.2 Objectives

By the end of the lesson you should be able to:

1. Define a river basin.
2. Describe the processes within a river basin.
3. Explain the catchment balance concept.

5.3 Stream flow

Stream flow comprises the gravity movement of water in channels of various sizes. The unit of measurement is the cumecs (m^3/s). It is often referred to as river discharge, runoff or catchment yield. Runoff may be expressed as a depth equivalent over a catchment in a specified time period; day, a month or a year. The sources and components of runoff include, direct precipitation, overland flow, shallow sub-surface flow and deep sub-surface flow.

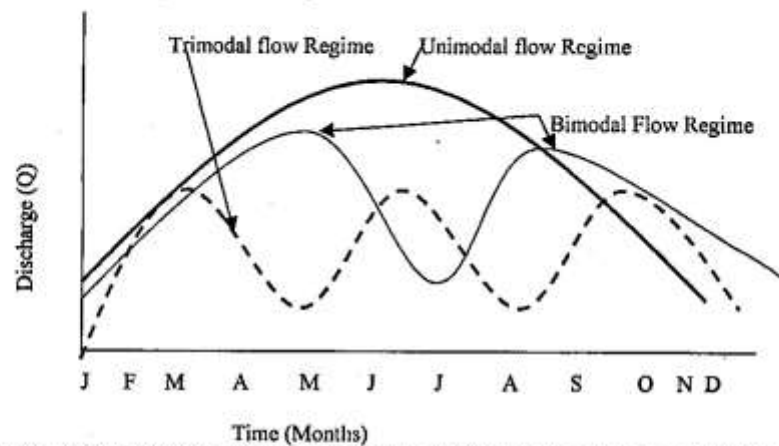
5.4 Overland Flow

Overland flow is water that fails to infiltrate the surface and it travels over the ground surface to a stream channel. Throughflow is water that has infiltrated and moves laterally through the upper soil horizons toward the stream channels above the main ground water level. Similar terms to throughflow are interflow or surface storm flow.

5.5 River Regime

River regimes are those variations of river flow throughout the year in which a simple distinction may be made between one period of high water level and one of low water level or movement. Such regimes may result from a number of factors including; climatic factors; temperature and precipitation characteristics as well as land use. Figure 9 shows of typical river regimes.

Figure 9: Examples of River regimes



A regime depicts characteristic seasonal patterns of river flow at any site established usually over a climatic period. The annual runoff variability reflects the variability of precipitation within the basin. There is marked contrast in rainfall variability between humid and dry areas and tropical and subtropical areas.

5.6 Classification of rivers

Rivers can be classified as perennial or ephemeral. Perennial rivers flow at all times some with low seasonal fluctuations as is typical of those in humid areas and those with high seasonal fluctuations as those in Arid and semi-arid areas.

Ephemeral rivers are those that flow intermittently and can be seasonal or episodic and have no base flow. The seasonal rivers dry up frequently in dry season and the episodic rivers are dry most of the year, but respond to rainfall events during the wet season. These rivers are typical of small basins in semi-arid and arid regions where rainfall is erratic but with high intensity; soils are impermeable and no contribution from ground water. The response to rainfall events is usually rapid yielding flash floods with high erosion potential.

5.7 Floods

Floods are the high-flows at site usually occurring during high precipitation periods. In most cases it overflows the riverbanks and inundates low-lying areas or river valleys and settlements. Flood frequency analyses are concerned with estimating the characteristics of annual floods and an understanding of this useful in the construction of hydrologic structures such as bridges.

5.8 Catchments Water Balance

The water balance can be developed to account for hydrologic components. This, for a catchment, is in the form of a simplified equation depicting the hydrologic components of Input, Output and Storage. The hydrologic equation then takes the form:

$$P = Q + AE + DSM + DGW$$

Where

P= Precipitation

Q= Runoff

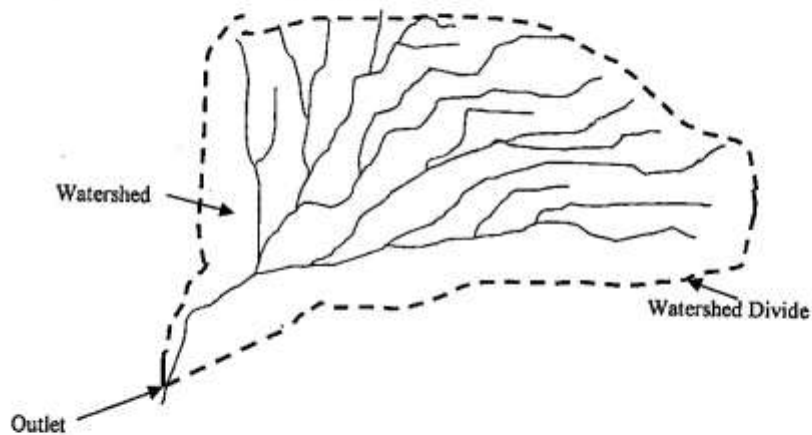
AE= Actual evapo-transpiration

DSM= Change in soil moisture storage

DGW= Change in ground water storage

All units are uniform usually in mm and for a specified period, usually one year. The components pertain to a catchment or watershed as indicated in the Figure 10.

Figure 10: A watershed or catchment



Important for the watershed is its time of concentration, which may be defined as the time required by the water to reach the outlet from the furthest point of watershed.

5.9 Summary

In this lesson, the concept of the water balance has been discussed. The concept is applicable for any scale including continental, regional, field or even a plot. The ideal hydrologic unit for considering the water balance is the watershed, river basin or catchment. These terms are used interchangeably. A river basin is defined as an area that drains to a single outlet. In such an area usually defined by topographic divide the dynamics of flow are presented to include inputs, outputs and storages and represent the law of continuity (mass balance) upon which the water cycle is based. Within the basin processes such as stream flow, overland flow, flooding among others takes place.

5.10 Revision Questions

1. (a) Explain the distinguishing characteristics of a river basin.
(b) Illustrate why it is necessary to understand drainage basin characteristics.
2. Identify and explain any three hydrologic processes within a river basin.
3. Explain the role of catchment water balance in drainage planning and development.
4. Using an illustration, explain what is meant by river regimes.

5.11 Definition of Terms

Water balance – This is a reflection of inflows and outflows into a basin. The difference between inflow and outflows represents storage in the basin.

A drainage divide – This is the line of separation between runoff that descends in the direction of the drainage in question and that, which goes toward an adjacent basin.

5.12 Further Reading

1. Shaw, E. M. (1998). **Hydrology in Practice**. Prentice Hall, London.
2. McNight, H. (1995). **Physical Geography, A Landscape Appreciation**. Prentice Hall, Loss Angeles.
3. Doherty, A., and McDonalty, M.(1999). **River Basin Management**. Hodder & Stoughton, London.
4. Whittow J, B, (1984) **The Penguin Dictionary of Physical Geography**. Allen Lane, London.

PART II: THE LITHOSPHERE

LESSON 6: GEOMORPHOLOGICAL PROCESSES

6.1 Introduction

The lithosphere is the solid part of the physical environment. The lithosphere was introduced in Physical Geography I. Mainly examined were endogenic processes that help to build the lithosphere. In the following series of lectures, we will mostly examine exogenic processes operating on the earth's surface and being driven by solar radiation and force of gravity.

These Geomorphological processes include weathering, erosion, and mass movement among others. The processes are brought about by geomorphological agents such as water, wind and glacier.

6.2 Objectives

By the end of the topic you should be able to:

- (i) Evaluate the role of endogenic and exogenic factors in rock decay.
- (ii) Describe different types of weathering processes
- (iii) Examine processes of weathering in the world's climatic regions
- (iv) Assess the place of weathering in man's decisions.

6.3 Weathering

This is one of the exogenic processes operating on the earth's surface. As a process of **denudation**, it wears away or destroys initial landscape built by mountain building processes. In this topic, the process is defined and factors influencing rock decay examined. Types of weathering processes and role of climate in rock disintegration have been discussed. Included, is the role of weathering in man's activities.

Weathering involves the breakdown and decomposition of rocks at or near the earth's surface by physical and chemical processes. The broken or disintegrated materials form a layer of debris.

Weathered materials (debris) are removed by various agents of transport such as water, ice, wind and mass movement.

Through the process of weathering, rocks adjust to their environment. The process of weathering is an **equilibrium process** where new rocks adjust to new surrounding conditions relatively different from their place of origin. Through weathering rocks adjust to the new physico-chemical conditions.

6.4 Factors in Rock Weathering

These are the forces involved in rock decay or disintegration. They are broadly classified into two that is **endogenic** and **exogenic** factors. These forces affect the nature, rate and character of weathering.

Endogenic factors involve the **internal characteristics** of the rock. These forces operate within the rock being weathered. They include; **mineral composition, rock structure and rock texture**.

Rock structure involves the presence of joints, faults, bedding planes and cracks or crevasses within rocks. The presence or absence of rock features will influence the rate of rock decay. Through these lines of weakness, water and dissolved substances easily attack the rock leading to disintegration. The surface area exposed to agents of weathering is in a well jointed rock is far greater than that exposed in a massive rock. The joints and fissures allow the removal of weathered materials exposing more rock area to weathering. This prevents the water from coming into equilibrium with the rock.

Rock texture too influences nature and rate of weathering. Texture refers to the **particle size** of the rocks. Small particles present a larger surface area for weathering. Fine-grained rock is weathered faster than a coarse grained rock.

Rock composition

It is the mineral grains of a rock that decay. The chemical composition of the rock minerals and the proportion of the different minerals influence the overall nature of weathering processes. Some minerals are quickly weathered such as **rock salt** that is removed in solution. Different rock minerals are susceptible to different chemicals weathering processes. Calcite rocks are affected by carbonation while rock salt is affected by solution.

Exogenic factors are causes of rock decay but are external to the rock. They are environmental factors. They include; climate, vegetation, slope and time.

Climate determines the availability of water responsible for chemical weathering, atmospheric pressure and soil temperature. Water is required for both physical and chemical weathering. High temperatures accelerate the rate of physical and chemical processes.

Decaying **vegetation** releases organic acids that aid in some types of chemical decay. Vegetation cover protects rock surfaces shielding them from wash processes and further decomposition. Open vegetation of Savanna type exposes rocks to both physical and chemical weathering processes.

Areas with **steep slopes** (steep gradient) undergo further rock decay as agents of transportation such as wind, water, and ice together with mass movement quickly remove

debris. On gentle slopes debris form a thick mantle protecting underlying rock surfaces and consequently reducing the rate of weathering.

Time is an external factor-influencing rate of weathering. The longer the time a rock has been exposed to processes of disintegration, the higher the rates of weathering. Recently formed rocks (young rocks) have had less exposure to weathering.

6.5 Types of Weathering

Two types of weathering have been identified – **physical and chemical weathering**. Physical weathering involves rock disintegration entirely by mechanical means. The breakdown is brought about by stresses acting upon or within the rock. The chemical nature of the resulting fragments remains the same.

In chemical weathering, rock minerals are decomposed by agents such as water, oxygen, carbon dioxide and organic acids.

A third type of weathering known as **biotic weathering** has also been identified. It is brought about by organic factors such as splitting of rocks by tree roots (physical processes) and rotting of rocks (chemical weathering) such as limestone due to bird droppings. Critics of the third classification arguably say that biotic weathering is embraced in the first two forms of weathering.

6.5.1 Physical Weathering

It is also known as mechanical weathering. This is the breakdown of rocks without altering their chemical composition. Some of the physical weathering processes include:

Thermal weathering – this results from the expansion of rocks upon heating and contracting upon cooling. This process is common in desert environments with great extremes in air temperature and little or no vegetation to protect the rock. Due to high temperatures by day and low temperatures by night, differential expansion does take place. The differential expansion may cause flakes of rock to crack away from the surface, producing the effect of **onion weathering or exfoliation**. Well-jointed rocks are most affected by exfoliation where curved rock peels off. When exfoliation is massive in well-jointed rocks, features known as **exfoliation domes** are formed.

Freeze-thaw

In this process, water is the main agent of weathering especially when it freezes. The process is common in high altitude and latitude areas. Ice crystal growth is encouraged in the joints or cracks. Under suitable conditions of ample moisture supply, ice crystal growth is encouraged. The increase in volume of the water as it freezes set up stresses that gradually and finally causes rock disintegration. Repeated freezing and thawing increases the degree of rock disintegration.

6.5.2 Biological weathering – plant roots have been responsible for mechanical weathering. Roots develop down the joints or bedding planes exerting large stresses on

the surrounding rock. This may, though to a limited extent, weaken and disintegrate a rock.

Wetting and drying – As a rock is wetted, water traps air bubbles within the rock. When the wetting is rapid, the water may explode causing ruptures in the rock. Hydration may occur. When the water retreats or dries up, expansion occurs widening the cracks/joints/crevices. The repeated wetting and drying will cause disintegration.

Pressure release (dilation) - When overlying materials covers rocks, they experience confining pressure. If the materials are removed for instance during quarrying, the confining pressure is removed, the rock expands, cracks develop and this weakens the rock. Water may enter the cracks and cause more disintegration.

Crystal growth – Most common in desert and coastal areas. Saline water evaporates, crystals of salt form and collect in the cracks and joints. The growing crystals exert a lot of pressure in the cracks forcing them to widen. The crystals behave like those of freeze-thaw action. The rate of disintegration will also depend on type of salt dissolved in the water. However, some rocks are resistant to this type of weathering.

6.5.2 Chemical Weathering

This is a type of weathering involving a reaction between rock minerals, water and certain atmospheric gases such as oxygen and carbon dioxide. It also includes reactions between organic acids that develop from decaying vegetation and rock minerals. This type of rock decay is most active in areas of high temperature and high humidity. These conditions favour rapid weathering.

In chemical weathering, three general types of reactions can be identified. Those in which **ions** from the rock minerals are removed and enter the water; those in which **ions** from the water enter and combine with the mineral; those in which exchange of ions take place.

In chemical weathering, water charged with different gases penetrates the rock dissolving and carrying away certain minerals in solution. The original mineral compounds are thus transformed into new, perhaps delicate compounds, thus weakening the rock.

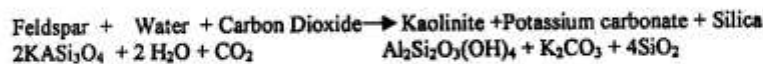
Some of the chemical processes of weathering include:

Solution

Some rock minerals are soluble in pure water for instance rock salt (NaCl) and are thus removed in solution. The acidity or alkalinity of the water is key in this process. Highly alkaline water removes Silica and Alumina from the rocks (Al_2O_3). In neutral conditions, Alumina is insoluble hence resistant to decay. Some rock substances enter into the solution through the action of reactants dissolved within the water further weakening of the rock.

Hydrolysis

This is one of the important chemical reactions occurring when free hydrogen or hydroxyl ions in the water enter into the mineral structure of the rock and create a new compound. In feldspar, hydrogen ions in water react with hydrogen ions in the rock. Hydrolysis of feldspar results in the formation of clay.



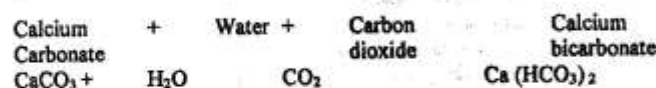
Some of the products of this reaction (Potassium carbonate) are removed in drainage waters. Carbon dioxide greatly accelerates the reaction. The Carbon Dioxide originates from free atmosphere or from respiration of plants and organisms.

Hydration

Some rocks absorb water and expand. In the process of volumetric changes, physical stresses are set up causing fracturing of the rock. Common type of hydration is conversion of iron oxides (hematite) to iron hydroxides (limonite).

Carbonation

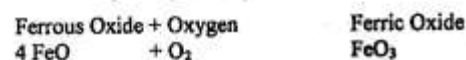
This is the process by which carbonic acid (a weak solution of carbon dioxide and water) changes calcium carbonate into calcium bicarbonate readily removed in solution by ground water.



Calcareous rocks mainly limestone and dolomite are weathered by this process.

Oxidation and reduction

Oxygen reacts or combines with rock minerals. Mainly affects rocks containing Iron (Fe). A rock, which has undergone this type of weathering usually, develops a yellow or brown crust. This new crust readily crumbles causing the rock to disintegrate.



6.6 Effect of Climate on Weathering

Type and rate of weathering vary from one climatic zone to another. In Equatorial climate characterized by high temperature, rainfall and humidity, higher rates of chemical weathering are constantly at work. Removal of weathered materials is low and a deep mantle accumulates on the surface forming deep soils.

In Tropical climate areas, there occurs alternate wet and dry seasons. Evaporation is high during the dry season and leaching is highly pronounced during the wet season. Laterisation (formation of subsurface crust) occurs. Chemical weathering by oxidation

takes place during the wet season. Physical weathering is a characteristic of the dry season.

In **arid and semi-arid areas** thermal weathering resulting differential expansion and contraction is a common process of decay. This is due to high and low temperature by day and night respectively.

Mid-latitude and polar regions are affected by mechanical weathering by frost action. In calcareous rock areas, chemical weathering by solution takes place. In high latitude areas, always covered by snow and ice, weathering may be limited. Regions under glacier and margins of ice sheets may experience chemical weathering.

6.7 Effects of Weathering

The consequences of rock decay can be viewed on two levels; first on its effect on landform formation and second, its effect on man.

(a) Landforms of weathering

Weathering alters the morphology of the zone within which it operates. Formation of **tors** in Yorkshire, England and Western Kenya are products of weathering. Tors consist of irregular piles or masses of rocks of perched on hilltops or on plateaus hedges.

They are formed when granite and sandstone rocks are weathered. One school of thought argues that tors are entirely products of chemical weathering and exhumation in tropical environments. The second school argues that tors result from frost shattering in polar conditions. In well jointed rocks, disintegration by freeze-thaw and associated mass movement of debris down slope, result to **tor** formation.

Tors may also result from pressure-release following fluvial incision. In conclusion, tors are a product of many processes either working independently or in combination.

In limestone environments, most beautiful landforms of weathering are conspicuous. The process occurs along joints and bedding planes producing features such as **clints** and **grikes**. On the surfaces between joints, **weathering pits** are formed. In the same environment, due to carbonation **caves** also forms. These features are generally termed as karsts landscape.

6.8 Weathering and man

- a) Weathering processes have their own implications on man. **Bauxite**, a major input in **Aluminium smelting** in the world, is a product of hydrolysis of feldspar.
- b) **Kaolinite**, a product of hydrothermal weathering is useful in the **paper and ceramics industries**.
- c) For **agricultural economies**, weathering is a fundamental process in **soil formation**.
- d) Weathering also affects **engineering structures**. The weathered mantle is necessary to construct buildings. A firm foundation for buildings is required but

weathering may weaken the surface materials, particularly where weathered materials are deep. The creation of **stable foundations** may present a problem.

- e) **Road cutting and embankments** must also take due regard to the effects of weathering. Weathering of roadside rocks and mass movement affects roads hence intervention measures such as stabilization becomes necessary.
- f) The effects of **solution and hydrolysis**, though on a small scale, wear many buildings.

6.9 Summary

Weathering is one of the processes that wear away the earth's surface. The nature and type of weathering is affected by both endogenic and exogenic factors. The former deals with rock characteristics while the latter involves controls of weathering from the outside environment. Rock decay varies from one climatic region to another. However, in most cases the processes may occur in combination, with one being dominant. In one way or another, man's activities are dictated by weathering for instance agriculture, road construction, slope accidents and architecture.

6.10 Revision Questions

- (a) Explain why you think weathering is a disequilibrium process.
- (b) To what extent is the rate and nature of weathering dictated by endogenic and exogenic factors?
- (c) How does weathering influence man's decisions?

6.11 Definition of Key Terms

Weathering as an equilibrium process -- rock materials attempt to adjust to the new physico-chemical processes. Initially, materials are in disequilibrium but become stable with time unless prevailing conditions change.

Endogenic factors in rock decay -- these are factors within the rock such as structure, texture and composition.

Exogenic factors in rock disintegration -- These are factors external to the rock that influence the breakdown of rocks such as slope, climate and man.

Clints and grikes -- Weathering features in limestone rock areas.

6.12 Further Reading

1. Thomes, m. F. (1996). **Geomorphology in the Tropics**. John Wiley & Sons. New York.
2. Buckle, C. (1986). **Landforms in Africa**. Longman, London.
3. Briggs, D. & Smithson, P. (1995). **Fundamentals of Physical Geography**. Routledge, New York.
4. King C, A (1980) **Physical Geography**, Oxford Blackwell.
5. Holmes A. (1964) **Principles of Physical Geology**. Thomas Nelson & sons.

LESSON 7 EROSION AND RESULTANT LANDFORMS

7.1 Introduction

Erosion is one of the geomorphological processes responsible for landscape modification. Surface erosion may be by water, wind or glacier. Each erosional process leads to the formation of a number of landforms. In this lesson, the cycle of erosion as one of the models of landform development is explained. Water, wind and glacial erosional processes and resultant landforms are examined in detail.

7.2 Objectives

By the end of the topic you able to:

1. Describe and critique William Davis Morris' theory of landscape development.
2. Explain the process of water erosion and its effect on landscape development.
3. Describe the process of wind erosion and resultant landforms.
4. Explain the process of glacial erosion and resulting landforms.

7.3 The Cycle of Erosion in Landscape development

Landscape analysts have envisaged landform development in a number of ways. One theory of landform development is that of the **cycle of erosion** as put forward by William Davis Morris. According to this theory, landforms, just like organisms evolve through a number of stages. It borrows heavily from Charles Darwin's theory of evolution.

As per Morris's theory, evolution of landscape follows a systematic sequence of stages (cycles or phases) designated as **youth, maturity and old age**. The theory assumes an initial uplift of land as the starting point of the cycle of erosion. Landforms are gradually eroded through the stages mentioned to a final monotonous plain known as a **penplain** (peneplanation hypotheses).

In the **youthful stage**, Morris's landform development begins after the swift uplift of a landmass above sea level. Rivers begin to erode the initial surface draining directly to the sea as consequent streams attempt to erode down to sea level.

The rivers' erosive power is high causing vertical erosion. Streams are strong and erosion of landforms is at maximum. V-shaped valleys are common. Tributaries are few, valleys deep and narrow.

The **mature stage** is marked with decreasing relief. The streams are not cutting down as rapidly as in the youthful stage. Subsequent streams developing along lines of weakness join the early consequent streams. Obsequent streams that flow generally opposite to the consequents in turn join subsequent streams.

In the mature stage, valleys are generally broadened and flood plains begin to emerge. Meanders may also begin to develop along the rivers. Many of the streams at this stage are **at grade**. This means that their gradient is sufficiently steep to allow water flow without much erosion. Land becomes progressively less steep.

In the **old or senile stage**, little upland, if any, remains except for occasional inselbergs (monad nocks). A thick waste mantle that effectively prevents further erosion covers the landscape. Stream gradients are very gentle, deposition common and broad flood plains formed. Rock surfaces are buried below the thick mantle. Very wide meanders, ox-bow lakes and cut-offs develop. Tributaries are few and sluggish.

In the advanced final stage, a low land of faint relief is formed and this is termed as a **peneplain**. William Davis Morris assumes flattening of landscape to this final landform, the perfect slope.

Critique of Morris' Cycle of Erosion

Davis's ideas captured the imagination of many students and for a long period landscape development was explained in terms of youthful, mature and senile/old stages. An evaluation of Davis's theory of landform development has some of the following weaknesses.

That the Cycle of Erosion was based on the **humid tropics** of USA leaves us wondering whether the model would apply in dry areas, where water, the main agent of erosion suggested by Davis, is scanty. Davis attributed **deserts** to climatic accident.

That **vegetation** can exert a frictional effect on the cycle of erosion also deals it a blow. The peneplain may take longer to be achieved in areas of thick vegetation. Davis had neglected vegetation yet it appears to have an important effect on landscape development.

The evolution concept. The concept of landscape evolving like Charles Darwin's organism also raises many questions than answers. Does landscape development go through the stages mentioned i.e., youth, mature and senile/old stages? Landforms do not evolve along a set course towards an inevitable end form, the peneplain.

The cycle concept is also far from reality. According to Davis, landform development is in a form of a cycle. This implies that after a certain period of development, land will return to the starting point. In reality, despite forces of denudation, land will never go back to the starting point.

The Davisian model of landform development lacked measurements. It cannot therefore be subjected to statistical analysis to assess its objectivity. It is only based on generations about landform development.

The cycle of erosion also gave little emphasis to **structure, process and time**. For instance, some rocks (hard) may resist erosion to an extent that the achievement of a peneplain may be a nightmare. This is for instance witnessed in the presence of inselbergs and other erosional landforms.

The concept of time further deals a blow to the Davisian Cycle. How long will it take to achieve William's peneplain?

Lastly, endogenic processes such as faulting, folding and vulcanicity continuously modify the landscape. In fact they make the peneplain hard to achieve.

In conclusion, though the geomorphic cycle was seriously criticized and is now generally dismissed, no completely satisfactory substitute has been found. This theory of landscape development is still credited for it shows that landform development is a dynamic process. Landscapes are not static but continuously changing.

7.4 Channel processes

On the earth's surface, a bulk of water occupies hollows on the ground or flows in definite channels. Some of the channels are manmade though the many of them are natural. The channels are continuously being modified by erosion either vertical deepening the channel or lateral i.e., widening the channel.

Laminar flow

This flow occurs when stream flow takes place in a relatively smooth channel, with little or no mixing lateral mixing.

The movement is in relatively definite, relatively smooth channel and no transverse mixing.

It occurs only in very small depths and at low velocities causing little or no erosion in the river channel.

Turbulent flow

This is flow in a river channel characterized by lateral and vertical mixing of flow in the river channel. It increases with depth of water. Nearly all channel flows are turbulent.

Stream flow energy

Flow in a river channel may be classified according to energy. The energy of flow is a function of its depth and velocity. The energy may be minimum at one particular depth yet critical in another.

In upland headwaters where gradients are very steep, stream energy is also high. Erosion of land and the river courses is prevalent.

On middle reaches of the rivers, where gradients have greatly reduced, river energy also reduces.

Increase in channel flow will also lead to increased erosion.

Sediment Transport

Sediments are materials transported by stream flow. The materials are small particles of rock debris or organic material transported and or deposited by water. Sediment load may be subdivided into three components. These are bed load transported at the river bottom, suspended material and saltation where particles separate from bed load and bounce along in the flow. The amount of sediment transport will depend on stream energy.

On steep gradients, stream energy is high and sediments are transported. On gentle topography, river energy reduces and the sediments are deposited.

7.5 Surface Erosion by Wind

This section explains wind as one of the agents of erosion. It examines processes of wind erosion and resultant landforms. The influences of Aeolian processes on man are also highlighted.

Wind as an agent of Earth Sculpture

Wind is one of the agents of denudation. Its effect is most significant in dry areas (deserts). Almost 10% of the world's land surface is hot deserts. It plays a vital role in landscape development. A further 10% of the world continents is polar deserts (cold deserts).

Wind activity is not only confined to desert areas. In many semi-arid and temperate areas, the effects of wind erosion are of great significance. Wind erosion affects cultivated areas. Man's attempts to the land for cropping and his repeated tillage of the soil, have produced unprotected, powdery surfaces, susceptible to the erosive power of the wind.

Processes of Wind Erosion (Aeolian processes)

Wind erosion involves two main processes; deflation and abrasion. **Deflation** is the removal of soft loose materials such as weathered debris, by lifting and rolling action.

Abrasion on the other hand, is erosion whereby materials carried by wind are used as tools for erosion. It involves the blasting action of wind by blown sand. The process of abrasion is very effective during sand storms whereby it polishes hard rocks and undercuts weak rocks.

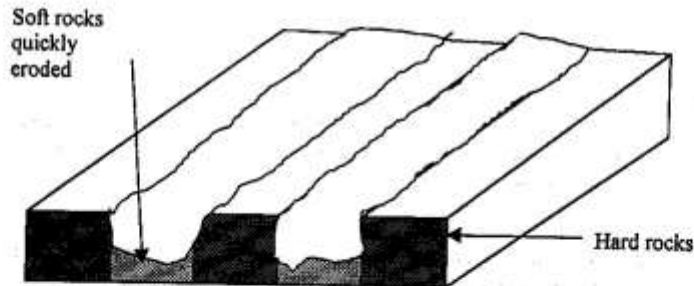
Wind erosion by abrasion decreases with height. This is because wind can lift very few materials above 50 cm. Most abrasion is therefore, at ground level.

Wind Erosion Landforms (Aeolian landforms)

Wind erosion by deflation, abrasion and attrition has resulted to a number of landforms among them yardangs rock pedestals, zeugens, blowholes, inselbergs among others.

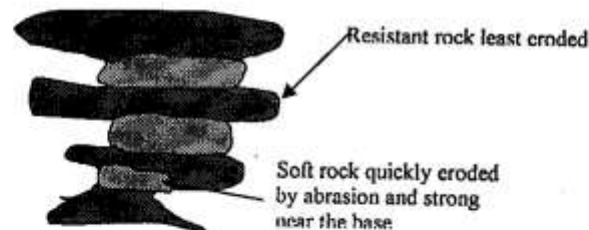
Yardangs are erosional features formed in areas of alternating hard and soft rocks arranged vertically or inclined. These extend parallel to the direction of prevailing wind. Wind erosion by abrasion wears away soft rocks. What is left is a features characterized by ridges (formed from resistant rocks) and furrows (areas formerly occupied by soft rocks). The height of yardangs may be up to 6 m high and 37 m wide.

Figure 11: The Yardang



Rock pedestals are formed in areas whereby alternating resistant and less resistant rocks arranged horizontally. The rock mass is attacked by abrasion. Materials carried by wind erode the rock mass more at the base than the top. Erosive power of wind by abrasion decreases with height. The feature formed is eroded most near the base that is undercutting is pronounced near the base.

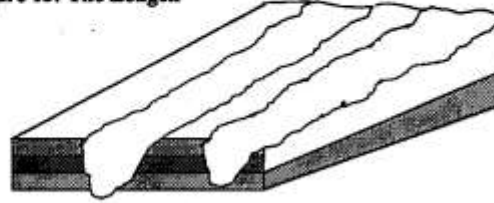
Figure 12: The Rock Pedestal



Zeugens are also formed in areas with alternating soft and hard rocks lying horizontally. The hard rock is underlain by a soft rock. Through the process of weathering (physical weathering), the hard rock is broken by abrasion. Once the soft

rock layer is reached, it is quickly eroded by abrasion. The resulting feature is characterized ridges and furrows where the hard rock is left upstanding on the soft rocks. The feature stands above the surrounding areas. Zeugens are later undercut and worn away.

Figure 13: The Zeugen



Blowouts

They are also known as pans or deflation hollows. They are depressions formed in areas with outcrops of weak or less resistant rocks. Wind erosion by deflation attacks the weak exposed rock and the place of occurrence is deepened.

The depression formed is further weakened by chemical weathering as the hollows formed retain moisture. The process is even faster in areas already affected by faulting. The materials weakened by weathering are easily blown away by wind deepening the hollow even further.

With repeated deflation, the water table is reached and wind can no longer be able to lift out wet particles. A shallow lake may then form on the floor of the depression. This is known as a blow out or oasis. This is the main source of water for desert inhabitants. Some are several kilometers wide while others are small. The Qattara Depression in Egypt is an example. There are also several pans near Botswana. The pans may be dry but hold water for several months once it has rained.

Figure 13: Deflation Hollow



In uniform (homogeneous) rock areas, undercutting by wind abrasion leads to the formation of **mushroom blocks**. The feature is undercut at the base hence narrow. Its top is broad due to reduced abrasion and is polished. Individual faceted or smoothened rock structures that fall from rock pedestals and mushroom blocks are termed as **ventifacts**.

The work of wind is not only limited to erosion but also transportation and deposition. In the latter, it leads to the formation of dunes i.e., **barchans, seif and transverse dunes**. It also leads to formation of very large features known as **draas** as well deposit fine materials known as **loess**, useful as fertile soil in areas of deposition.

Aeolian processes and man

- (a) Soil erosion by wind – wind is an agent of erosion. It blows away soils in form of dust. Fields not only loose soil but the nutrients associated with the soil. This will lead to reduced yields.
- (b) Wind erosion and deposition leads to the formation of loess, a very rich agricultural soil.
- (c) There has been increased research in soil science on how to conserve soil and therefore remove the negative effects of wind erosion. Scientists and agriculturalists have been endeavouring to understand more clearly what controls soil erosion by wind.
- (d) Aeolian processes are partly responsible for desertification. In the more recent time a general question is being asked: are deserts advancing? Some observers certainly think so. In the Sudan, deserts are known to have advanced between 90-100m to the south. The rates of advancement are however difficult to measure. Aeolian processes, aggravated by man's activities such as overgrazing, have been held responsible for desertification.
- (e) Pans are sources of water for desert inhabitants. These are formed through Aeolian processes especially deflation. The Qattara depression in Egypt is such a good example.

7.6 Glacial and Periglacial Systems

This topic introduces glaciation with the glacial mass budget. Processes of glacial erosion and resultant landforms are explained. Landforms of glacial deposition are also surveyed. The effects of glacial and periglacial on man are examined.

The Process Glaciation

Ice covers 10% of the world's land surface. It provides some of the very hostile and unexplored areas of the earth's landscape. In Greenland, Arctic and Antarctic, uninterrupted masses of ice are observed.

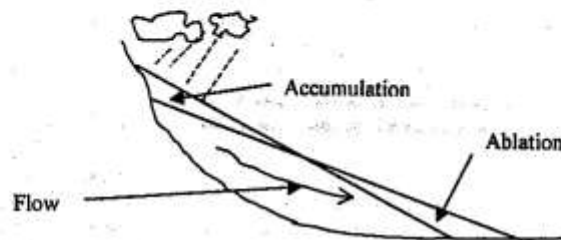
Ice sheets and glacier in particular is a major agent of earth sculpture. They aid in erosion, transportation and deposition of materials on the earth's surface.

The Glacial Mass Budget

The ice budget of a glacial system has inputs and outputs. In the upper part of a glacier, net accumulation takes place. In the lower zone, ablation (loss of ice in summer by sublimation – change from solid state to gaseous state and melting takes place.

An equilibrium line occurs where ablation and accumulation are in balance. Glacial flow maintains the balance either side of the equilibrium line and maintains the equilibrium profile.

Figure 15: The Ice Budget of a Glacial System.



Glacial Erosion

Glacier is one of the agents of erosion on the earth's surface especially in high latitudes and altitudes. Processes of glacial erosion include:

- (a) **Plucking** – this is the quarrying process of glacier. Rock materials on the way of glacier get frozen and are carried away as glacier moves. Plucking however depends on a number of factors including but not limited to; resistance of underlying rock, glacial speed, thickness and weight of glacier and amount of rock debris.

- (b) **Abrasion** – A process of erosion whereby materials carried by glacier are used as tools for erosion. Glacial materials scrape the floor and sides of the valley thus eroding them.

Landforms of Glacial Erosion

Features of glacial erosion are most conspicuous in highland areas, though depositional features may also be found. Erosional features include but are not limited to: cirques, arêtes, pyramidal peaks/horns, glacial troughs and fiords.

(I) On the Highlands.

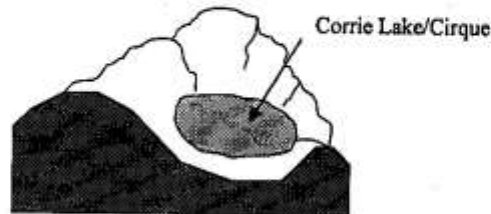
(a) Cirques

These are steep sided rock basin like features formed by glacial erosion on the highlands or mountains and valley heads. They are also known as corrie lakes. They develop through the accumulation of semi-permanent snow in patches. Due to freeze-thaw action, the snow gradually excavates a shallow depression known as a nivation hollow.

The initial hollow is deepened by abrasion while plucking steepens the walls. Repeated abrasion and plucking as well as removal of weathered debris from the depression leads to the formation of over-deepened basin. The cirque is armchair shaped.

Melt water helps to wash away disintegrated debris.

Figure 16: Cirque



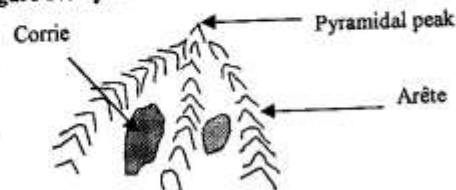
(b) Arête

An arête is a narrow ridge between adjacent corries. It is a steep sided rocky ridge and separates two cirques. It forms when there is back wall recession of the cirques into the mountainside.

(c) The Horn

This is also known as the pyramidal peak. It is a steep sided peak surrounded by a cirque. It is formed due to recession of arêtes. In fact, from the top of a pyramidal peak, arêtes radiate in different directions.

Figure 17: Pyramidal Peak



(d) Glacial Trough

These are often broadly u-shaped cross-sections formed both by vertical and lateral glacial erosion. The ground and sub-glacial moraines scrub the walls. Vertical and lateral erosion are almost equal in magnitude. This kind of erosion leads to the formation of a broad, flat-bottomed, steep-sided feature with a u-shaped cross-section, the glacial trough.

A glacial trough is also formed when glacier occupies a pre-existing river valley. Lateral and vertical glacial erosion modifies this initial river valley into a glacial trough.

(e) Hanging Valley

The degree of valley erosion is a function, among other things, of ice depth with the result that large valleys tend to be deepened more than small ones. Tributary valleys, which carry only small glaciers, may be left as hanging valleys.

Alternatively, hanging valleys are formed where two or more glaciers converge. If one of them has more erosive power, it is deepened more than its counter part. The weaker glacier is left at a higher level forming a hanging valley. Streams from hanging valley plunge into the glacial trough forming a waterfall.

(f) Rock Basins

A rock basin is a feature that forms in the glacial trough. Locally increased abrasion or excavation of softer rocks in the valley floor produces a rock basin. They vary from a few metres to many kilometres in length. When ice has melted, they form lakes known as ribbon lakes.

A rock basin is also a product of differential erosion. It forms where two glaciers meet. With the convergence of two glaciers, erosion will be greater. A glacial hollow is formed and if melt water accumulate, the ribbon or finger lake forms.

(g) Fiord

This is a deep, narrow arm of the sea with parallel, steep sided walls. It is a submerged glaciated valley. The depth of fiord water may exceed 1000m. Glaciers greatly over-deepening coastal valleys formed fiords. Their greater depth emanated from the ability of glacier to erode well below sea level.

(II) Features of Glacial Erosion on the Lowlands

Erosive work of glacial is not only limited to the highlands. On the lowlands, distinct features are formed just as on the highlands. Some of the features include but are not limited to:

(a) Glacial Lakes/Ponds

In areas of soft rocks, moving ice by plucking and abrasion, scoop out and transport away materials leaving behind depressions. These may later fill with water forming glacial lakes and ponds.

(b) Roche Moutonnee

These are resistant rock masses rising above plains in glaciated areas. They vary in size ranging from small outcrops to quite large hills. Their upstream sides have been smoothed by ice, but the downstream side is steep and rugged due to plucking.

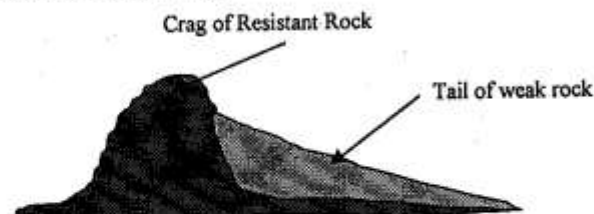
Figure 18: Rock Moutonnee



(c) Crag and Tail

In some areas, resistant rock outcrops obstruct ice movement protecting weaker rocks on the down streamside from erosion. The resultant effect is a feature with an elongated 'tail' of weak rocks, known as a **crag and tail**. In some instances, the tail may be constructed from deposited glacial till.

Figure 19: Crag and Tail



Landforms of Glacial Deposition

The stripped and scoured relief of glaciated uplands demonstrates the massive erosive power of glacial ice. However, this is only half the story of glacial impact on the landscape. All the debris scraped, chipped and plucked off the surfaces are available for deposition.

Some of the features of glacial erosion include but are not limited to; **tills, drumlins, terminal moraines, eskers, kames, kame terraces and outwash plains.**

(a) Till plains

This is a feature of monotonous relief, formed when former valleys and hills are buried by a thick layer of materials (till) beneath the ice sheet. Most till plains are a result of accumulated deposition during several ice advances.

They may have high clay content. Boulders till produce a landscape characterised by small hills and small depressions. When melt water collects in such depressions, kettle lakes are formed.

(b) Drumlins

A drumlin is a low, rounded elongated hill. It may be up to 1 km long and 30 m high. Most drumlins have somewhat steeper upstream ends consisting entirely of till. Some however, have a rock core.

Formation of drumlins has been a subject of debate. One school argues that they result from differential deposition while another school is of the idea that they result from the erosion of till materials. Despite the debate, modern ideas suggest that drumlins are a product of particles migrating into areas of low ice pressure such as cavities. The gradual closure of the cavities by internal deformation of the ice, account for drumlin formation.

(c) Terminal Moraine

This is an irregular mound or ridge of glacial deposits. It is formed by extensive deposition along the edge of an ice sheet and may be more than 50 m high. Many are composed of entirely of till. Their outer sides may be composed of a profile of poorly sorted fluvio-glacial deposits.

(d) Esker

This is a long winding, steep-sided ridge of sand and gravel. It may extend over many kilometres and rise as high as 30 m. Most of them are ice contact features and are formed by streams flowing beneath or within the ice. Formed from fluvio-glacial materials deposited by melt water.

Eskers develop where streams can maintain permanent sub-glacial tunnels. The under flows at high hydrostatic pressure carrying a large load, much of which may be deposited within the tunnel. When the ice melts, the gravel streambeds remain.

Eskers are similar in form to terminal moraine ridges, but they differ in being stratified and lying at right angles to any adjacent moraine ridge.

(e) Kame

This is also a landform resulting from fluvio-glacial materials. It is an irregular mound or hill of sand and gravel. Kames vary in shape and size. They may occur in isolation or in groups. They are formed in, on or at the margin of stagnant ice. Most kames show evidence of the collapse and slumping that occurred when ice melted.

(f) Kame Terrace

This is a discontinuous embankment along a valley side or upland slope. It is formed by the deposition of sands and gravels in narrow lakes held between the ice front and an adjacent upland.

Once the ice melts, the ice contact edge of the deposit will slump down. Kame terraces are similar in form to lateral moraine ridges.

(f) Outwash Plain

This is a wide gently sloping plain of gravels and fine sand. Deposits on outwash plains may be over 50 km thick. May hold small lakes. They are a result of enormous volume of melt water spreading materials in great fans beyond the ice front.

Braided streams drop coarse gravels first, while fine sands and clays are carried further away.

Periglacial Landforms

These are landforms that develop in high altitude and latitude areas under a climate characterized by alternate freezing and thawing.

In East Africa, periglacial processes operate on high mountains at altitudes where near-glacial conditions are dominant such as Ruwenzori and Mount Kenya. In these areas the ground remains permanently frozen except for the upper 1 or 2 metres. The permanently frozen sub-soil is known as permafrost.

In periglacial areas solifluction is a very important process. Deposits of solifluction gravels affect the landscape. They tend to conceal underlying irregularities in the relief.

Glacial and periglacial landforms and man

- a) Some till and outwash plains form part of the world's rich farming areas. The Prairies of Canada are an example.
- b) Some of the glacial lakes for instance the Great Lakes and St. Lawrence Sea Way form a natural route ways and sources of food.
- c) Some features resulting from glaciation attract tourists, a major source of income for many countries. Deep glaciated lakes in Switzerland are good

tourist resorts. During winter season sports such as skiing, skating among others are common.

- d) Waterfalls resulting from hanging valleys provide suitable sites for hydroelectric power generation.
- e) During the summer season, glaciated regions of the Alps Mountains in Southern Europe provide rich pasturelands.
- f) Erosion by ice sheets exposed minerals such as iron, cobalt, lead and copper in Canada.
- g) Fiords provide good fishing grounds as well as good natural harbours.
- h) However, fiords are inaccessible from the mainland.
- i) Some features of glaciation are a bother for instance, boulder clay deposits interfere with drainage by creating marshy landscapes.

Some outwash plains contain infertile sands that give rise to extensive areas of wasteland.

7.7 Summary

Landscapes are dynamic. They change over time. One of the theories of landscape development is that of William Davis Morris. Landscapes undergo initial uplift and go through stages; youth, mature and senile stages. A faint relief, the peneplain is finally formed.

In the process of landscape development, water is one of the agents. Water by rivers is an example. Wind is another agent of earth sculpture. It erodes by deflation and abrasion. Features of wind erosion include; yardangs, zeugens, blow outs, Inselbergs and rock pedestals. Some features of wind erosion are sources of water for instance blowouts provide desert inhabitants with water. Desertification is aggravated by the action of wind.

Glacier is one of the main agents of erosion and deposition on the earth's surface. It mainly erodes by plucking and abrasion. Erosional features are formed both on the highlands and the lowlands. On the highland some of the features include cirques, arête, pyramidal peaks among others. On the lowlands are roche moutonnée, crag and tail. Glacial depositional features include: tills, drumlins, kames and outwash plains. Effects of glacial and periglacial environments have been highlighted.

7.8 Revision Questions

1. Outline the Cycle of Erosion as postulated by William Davis Morris.
2. In your own view, do you think the theory is true?
3. Evaluate the role of structure, process and time in landscape development.
4. Explain why wind is a major agent of earth sculpture in arid and semi-arid lands.
5. What is the effect of wind erosion on landform development?
6. Evaluate the processes of wind erosion in desert areas.
7. Define desertification and explain the role of man in exacerbating the process.
8. Explain the role of plucking and abrasion in glacial erosion.
9. Highlight effect of glaciation on
 - i. Landform formation
 - ii. Human activities

7.9 Definition of Key words

- a) **Evolution Concept** – Development through sequential stages one after the other. The stages youth, mature and old are characteristic of evolution.
- b) **Structure** – Features on landscape have joints, faults and bedding planes. All these have an impact on development of landscapes.
- c) **Landscape processes** – That landforms are continuously being modified by endogenic (internal) and exogenic (external) processes.
- d) **Time** – an important aspect of landscape evolution. That landscapes change over time.
- e) **Grade** – A stage in landscape development when the gradient of the landscape is sufficiently steep to allow water to flow without much erosion.
- f) **Aeolian process** – process of wind erosion. Leads to the formation of a number of landforms.
- g) **Wind Deflation** – process of wind erosion whereby materials are lifted and rolled away by wind.
- h) **Wind Abrasion** – process of wind erosion whereby materials carried by wind are used as tools for erosion.
- i) **Blow out** – a place of water in desert areas (oasis). Natural source of water in desert areas.
- j) **Snow** – form of precipitation formed when temperatures fall below freezing point.
- k) **Contains water and tiny white crystals.**
- l) **Ice sheet** – This is a continuous mass of ice covering a wide area. Common in high
- m) **Latitude areas** such as Arctic, Antarctic and Greenland.
- n) **Glacier** – This is a mass of ice moving within a valley. Glaciers are common in both
- o) **high altitude and latitude areas.**
- p) **Iceberg** – a large mass of ice partially submerged in water, a phenomenon of high
- Latitudes seas/oceans or waters.
- q) **Ice cap** – Small masses of ice or glaciers covering plateau or mountaintops e.g., Kibo on Mt. Kilimanjaro.
- r) **Snow line** – This is there is permanent snow cover. It the dividing line between the areas seasonally occupied by ice and that covered by permanent snow.
- s) **Pleistocene** – This a period when glacial coverage and activity was restricted to the highlands such the East African Highlands, Ethiopian Highlands, the atlas, among others. Pleistocene ice age ended about 12000 years ago, having lasted for over 2 million years.
- t) **Periglacial** – ice conditions at and beyond glacial margins characterized by alternate freezing and thawing in response to temperature changes.
- u) **Moraine** – These are the materials (clays, sand, gravel and boulders) carried by glacier. The materials are also known as *Till*. The moraine may ground moraine (materials) underneath glacier, lateral moraine i.e., materials along the sides of glacier, medial moraine i.e., materials on the surface but in the center of a glacial mass. Terminal moraine is the material deposited by ice when it melts.

7.10 Further Reading

1. Pritchard, J. M. (1979). *Landforms and Landscapes in Africa*. Edward Arnold, London.
2. Thornburry, D. W. (1969). *Principles of Geomorphology*. John Wiley & Sons, New York.
3. Holmes, A. (1965). *Principles of Geology*. Thomas Nelson & Sons, London.
4. French, H. M. (1976). *The Periglacial Environment*. Longman, London.
5. Darrel & Weyman, V. (1977). *Landscape Processes*. George Allen & Unwin, London.
6. Small, R. J. (1992). *The Study of Landforms*. Cambridge University Press, New York.

LESSON 8 SLOPE PROCESSES AND MASS MOVEMENT

8.1 Introduction

This topic introduces learners' to slopes as important forms of our landscape. Slope concept, problem, system and stability are explained. Factors affecting slope stability and processes have been evaluated. The topic also explains theories of slope development. The influence of slope development and processes on man's activities is highlighted.

8.2 Objectives

By the end of the topic you should be able to:

1. Explain the magnitude of the slope problem.
2. Identify and describe slope processes.
3. Evaluate theories of slope development.
4. Explain how slopes affect man's activities.

8.3 The Slope Concept

Slopes are an important part of landscape. The landscape is actually a mosaic of slope units some very steep, other gentle and further still there are almost horizontal plains. Slopes involve material movement under force of gravity and there is also input in form of energy.

Slopes are important in practical terms as they exert the type of human activity on the slopes. They govern the type of agriculture to be carried out and the sustainability of the land for construction.

The use of machinery is affected by slope angle. The use of combine harvesters and tractors is limited by slope angle. In some countries, cereals cannot be cultivated on land steeper than 11° .

Some countries, however, extend cultivation to steep slopes above 15° (25° in Spain and 18.5° in Malaysia). In developing countries such as Kenya, due to population pressure, human activities may extend to steeper slopes.

There is therefore need to understand slope processes or dynamics for enhanced human activities; settlement and agriculture. Hazards associated with slopes (erosion and landslides) can also be mitigated if slope dynamics are well understood.

8.4 Magnitude of the Slope Problem

Slopes are one the most basic of all landforms that have commanded the attention of Physical geographer. Slopes directly or indirectly affect man's activities in agriculture, road and railway construction, house building, land drainage among others.

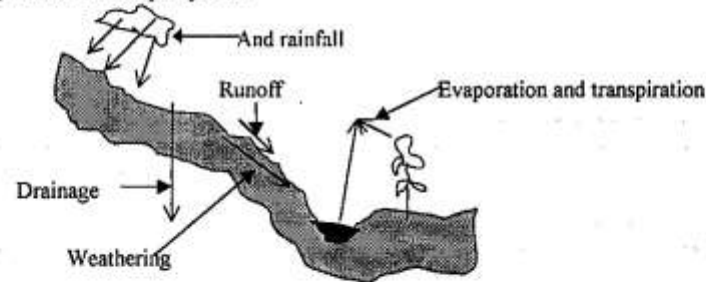
Slopes are associated with various kinds of instability that pose a serious threat to man. Understanding the mechanism of slope development constitutes a vital task for a geographer.

Slopes are not very slope but highly complex landforms. They vary according to locality, steepness and degree of active development. This reflects their influence on weathering, transport processes, rock type and climate.

8.5 Slope System

Slopes are dynamic that is changing over space and time. Like other systems, they are characterized by inputs and outputs. The inputs into the slope system include; solar radiation, precipitation, debris from weathering and dissolved substances.

Figure 20: The Slope System



Outputs from the system include; evaporation, percolation of water and dissolved substances into bedrock among others. At the foot of the slope, materials are removed from the slope system by streams, glacier and sea.

The inputs in the system initiate slope processes and consequently modify or change the nature of slope.

8.6 Slope Stability

This involves the resistance of slope to material movement down slope. If the materials easily move, the slope is said to be unstable and vice versa. Unstable slopes may collapse causing hazards such as landslides. Slope stability and material movement is a function of a number of factors.

- The angle of slope – The steeper the slope, the higher the degree of instability. This is associated with faster material movement down the slopes.
- The nature of slope material is also crucial. This is further dependent on the extent of saturation. Slopes are unstable where massive rocks overly weak rocks such as clay and shale. The more saturated the slope materials, the higher the degree of instability.

- (c) Climatic factors such as the amount and nature of rainfall influence slope stability. Comparatively heavy rainfall as well as freeze-thaw action encourages movement of materials leading to slope stability. Landslides are for instance most pronounced when rainfall is high.
- (d) The influence of vegetation on slope stability is also critical. Vegetation roots bind slope materials resisting down slope movement. Decaying vegetal materials form a mantle on slope surface that encourages infiltration (percolation) reducing runoff and material movement.
- (e) Tectonic factors such as earthquakes, tremors and vulcanicity often cause widespread slope instability and materials movement. This may even initiate hazards such as landslides.
- (f) Man's activities such as mining, building and herding of animals influence slope stability. More often, man's activities destabilize slopes causing slope hazards. Man's encroachment of steep slopes for cultivation is of significance.

8.7 Different Slope Processes

A number of processes take place on slopes leading to transfer of debris and other materials down slope under the influence of gravity. The processes vary considerably depending on slope materials (gravels, sands, soils, clay and hard rocks). The angle of slope also influences the down slope transfer of materials. On steeper slopes, there occur rapid processes of material movement. On gentle slopes, slow and perhaps imperceptible processes take place.

The following are some of the slope processes ranging from the very slow to very fast and hazardous processes.

(a) Soil creep

It is the movement of soil and fine materials down slope especially on very gentle slopes. The process is very slow and may be unnoticeable. Any process that disturbs soil such as alternate heating and cooling or wetting and drying will cause down slope movement. A sub-surface hardpan layer may act as a plane of medium of movement.

Soil creep can be evidenced in a number of places as it causes fences to bulge, tree trunks to bend in the direction of the creep, soil heaping at the foot of the slope and mounds of soils behind walls.

(b) Talus creep

This is the movement of angular waste rock materials of all sizes slowly down slope. In high altitudes and latitudes, melting of snow accelerates material movement. Ridges and mounds of angular boulders form down slope.

(c) Solifluction

Involves movement of saturated soil gravels and weathered rock. It is common on moderate slopes. It is mostly pronounced in mountain and cold climate areas where thawing causes a saturated surface layer to creep as a mass underlain by frozen ground.

Solifluction tends to fill hollows leveling the ground surface as saturated materials move down hill.

(d) Rock creep

Sometimes individual rock waste may move down slope at a slow speed. This is termed as rock creep. This happens when the underlying material or soil is clay.

(e) Mud flow

This is the movement of semi-liquid over-saturated soil down a moderate to steep slope. The movement of materials is through definite channels. The process is also known as soil flow. In arid and semi-arid (ASALs), mudflows occur whenever there is rain.

The saturated material flows down slope until, as the water is lost through seepage, the flow solidifies and comes to a halt. Mud flows may operate on very low slope angles, due to high moisture content. Only small gravitational stresses are necessary to initiate movement.

On steep slopes, the movement of materials can be very rapid. On relatively gentle slope, the movement is slow.

(f) Landslides

These are the fast to very fast down slope movements often very disastrous or hazardous. Large quantities of rock and soil suddenly move down hill or at the riverbank or along the roadside. This occurs normally after a down pour or continued rain. Some forms of landslides include:

(i) Rock slump

A down hill movement involving tearing away of rock material along a slope. The movement leads to back tilting of the surface and top of the beds. The materials tilt back forming a series of steps. Massive rocks overly weak rocks saturated by heavy rain. The process is common in over-steepened slopes such as scarps, cliffs and road cuttings.

(ii) Rock slide

It involves down slope movement of a mass of rock materials. The movement occurs over a bedding plane or fault plane. The materials actually slip over the bedding plane. Scars appear on hill slope.

(iii) Rock fall

This is a free fall of massive rocks or individual boulders down a very steep to vertical mountain slope. In well-jointed rocks, freeze-thaw may loosen rocks leading to rock fall or granular disintegration. The materials build up as a scree or talus slope at the foot of the slope.

8.8 Slope Development

On landscape, many different types of slopes can be detected with varying angles. Different processes give rise to different slope forms. Overtime, slopes may develop towards equilibrium forms. The question here is "*how do slopes evolve? What governs their evolution?*"

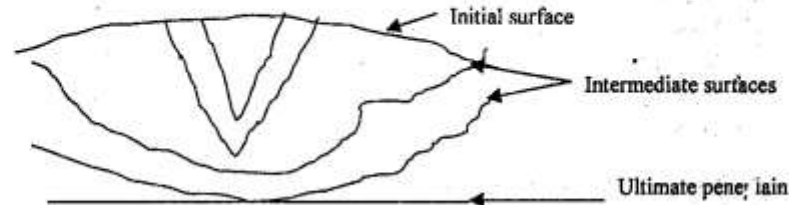
Three general processes of evolution have been put forward to explain slope development. These provide answers to the questions raised above. The theories of slope and general landscape development include: Slope decline, parallel retreat and slope replacement.

Slope decline: An American Geomorphologist William Davis Morris of the American School of Geomorphology postulated the concept of slope decline. This was part of his model or theory of landscape development.

He argued that landscapes undergo a cyclical evolution that begins with initial uplift progressing through processes of fluvial incision (water erosion) into a final flat surface, the peneplain (peneplanation).

During fluvial incision, relatively steep slopes are formed. As the river progresses, down cutting also ceases. Slopes of progressively lower angles are formed.

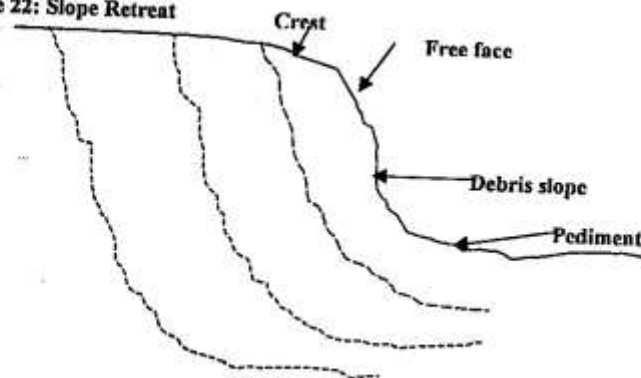
Figure 21: Sequence of Slope Decline according to Morris' Model



Parallel Retreat : L. C. King, a South African Geomorphologist, postulated this theory. The theory is known as King's model of slope evolution. According to King, slope development takes place by a process known as parallel retreat.

In this model, a slope is made up of a number of segments; **free face** (steep cliff) characterized by rock fall, **debris slope** (middle slope) characterized by slides and avalanches and the **pediment** (a gently sloping surface).

Figure 22: Slope Retreat



According to King, different processes characterize each of the segments. Soil creep is the main process on the crest. Landslides and rockfalls are active on the free face. On the debris slope, slides and avalanches take place. Sheet wash maintains the pediment as it washes the debris from other segment.

Slope development is controlled by removal of materials from the free face. As this takes place, debris slope also moves back, leading to parallel retreat of the whole slope. This further extends the pediment while hilltops diminish.

Late in the cycle of King's erosion, hills are left as isolated, steep sided relics, called inselbergs in Africa or buttes and mesas (as per size in North America).

King's hypothesis appears to be broadly valid in semi-arid areas and geologically stable environments of Southern Africa. The relevance King's model of slope and landscape development in other places or environments still remains controversial.

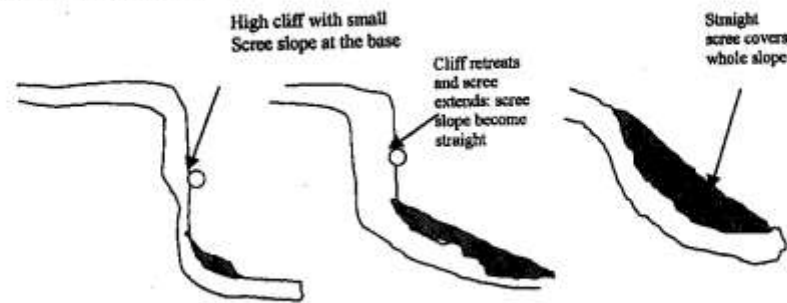
Slope Replacement: One of the weaknesses of King's model of slope development is the process by which the debris slope retreats in pace with the free-face. In the presence of active erosion, it is difficult to explain how this should happen. In reality, this does not always occur.

In place of parallel retreat, slope replacement takes place. As the cliff face retreats, the debris or scree extends until it totally covers the surface.

Initially, material falling from the cliff face accumulates as an ill-sorted pile of debris at the base. As the pile grows, debris falling onto it is able to bounce and roll away from the cliff. This results in the development of a low angle scree slope. The cliff continues to weather and retreat. More debris is trapped.

Finally, as the cliff height declines to zero, debris is released and movement takes place by sliding.

Figure 23: Process of Slope Replacement



8.9 Slopes and man

- On a small scale, collapse of embankments and localized landslips cause inconvenience and endanger life.
- On a larger scale, slope movements may be catastrophic. The landslides in Murang'a, Kenya are no exceptional.
- Lives and property are lost due to slope processes. Sometimes slope materials may move at speeds such as 400 per hour destroying everything on its way.
- Dams and other structures may collapse due slope processes. A collapse of a dam may cause unprecedented damage such as flooding.
- Hill slopes can be hazards and man is always responsible for increasing the hazard by mismanagement of the land.
- Steep slopes are unstable and affected by erosion (water and glacial).
- Collapse or caving in of roof mines may burry workers a life.

In conclusion there is need to understand how hill slopes function. This will enable us to cope up with or control environmental hazards associated with slopes. Resource investment and all architectural design on slope should be based on better understanding of slope processes.

8.10 Summary

Slopes are vital constituents of our landscape. Slopes are seen as a system with inputs and outputs. Slopes are affected by a number of factors such as slope angle, nature of its materials, tectonics and of course human activities. There are a wide range of processes on slopes ranging from those that are very slow and imperceptible to those that are very rapid and disastrous. Slope decline and parallel retreat theories explain slope development. Slopes affect man's activities such as building, agriculture and occurrence of hazards.

8.11 Revision Questions

1. Explain the magnitude of the slope problem?
2. What factors govern slope stability?
3. With illustrations, describe slope processes.
4. Why is the knowledge of slopes an important ingredient in Physical Geography?
5. Explain how slope nature and process affect man's utilization of slopes.

8.12 Definition of Key Terms

Debris – Materials resulting from rock decay.
Crest, free face and pediment – segments of slope according to Lester King.
Slope decline – process whereby slope angle decreases with time. Erosion consequently reduces and at grade, it is minimum and deposition occurs.
Slope stability – Strength of slope resistance to material movement down slope.
Slope system – Slopes changes over time and have inputs for instance precipitation and solar radiation. Slopes have outputs such as evaporation and drainage.

8.13 Further Readings

1. Selby, M. J. (1993). **Hill Slope Materials and Processes**. Oxford University Press, Oxford.
2. Clark, M. & Small, J. (1982). **Slopes and Weathering**. Cambridge University Press, Cambridge.
3. Schumm, S. A. & Mosley, M. P. (1973). **Slope Morphology**. Downsden & Hutchinson, Pennsylvania.
4. Strahler A. N and Sreahler A. H (1976) **Elements of Physical Geography**. John Wiley and Sons, New York.
5. Darrell and Weyman V (1977) **Landscape Processes**, George Allen and Unwin, London.

LESSON 9 COASTAL PROCESSES AND RESULTANT LANDFORMS

9.1 Introduction

In this topic, the nature of coasts is explained. Factors influencing coastal geomorphology are examined. Processes of wave erosion and resultant landforms are discussed. Included in this topic are landforms of wave deposition, types of coasts and effect of coastal systems on man.

9.2 Objectives

By the end of the topic you should be able to:

- a) Explain factors affecting coastal geomorphology.
- b) Evaluate erosional and depositional processes of wave erosion as well as resultant landforms on coasts
- c) Describe types of coasts
- d) Examine the effects of coastal systems on man.

9.3 Nature of Coasts

Coasts are a vital part of landscapes on the earth's surface. Developments along coasts in one way or another affect man's activities, more of the reason why we need to understand coastal dynamics. Waves and ocean currents that attack the rocks, eroding and depositing debris along the coastline, affect coastal landscapes.

Ocean currents transport materials over long distances, along the coasts and out of the sea. In the still dark depths of the oceans, the debris slowly accumulates. Some are trapped completely with no ready escape. They may lie there until plate movements and continental collisions force the ocean floor into mountain chains or the materials subsume into the mantle.

Landforms along the coasts are a product of wave erosion, transport and deposition. Erosion mainly affects highland coasts while deposition is the main phenomenon on lowland coasts.

The sea or ocean **erodes** the land mainly by the **hydraulic force** of the waves hitting the coast, combined with the **abrasive action** of the materials carried by the waves. The abrasive tools include sand, pebbles and large rocks. Wave erosion may also be in the form of **solution** whereby rocks that make up the coasts are soluble such as limestone, which is dissolved by the chemical action of the water. It is also important to note that materials (sand, pebbles and rocks) carried by waves hit against each other as they are hurled to and from the coast by the **swash** and **backwash**. This process is known as **attrition**. Through this process, the materials break and reduce in size.

Breaking waves brings about **coastal deposition**. On some coasts, wind is also an agent of deposition. Depositional landforms are partly built with materials eroded from the coast itself. Other materials are sediments brought by rivers.

9.4 Factors Influencing Coastal Morphology

Coastal shape is affected by a number of factors. These factors include but are not limited to:

- (a) Work of waves and tidal currents. Waves are the main agent of marine erosion, transport and deposition. Waves are undulations on the surface of the sea or ocean water. They are caused by wind blowing across the water. The wave movement to the coast is known as **swash** while the retreat of wave water is known as **backwash**. The **swash** causes erosion on raised coasts while the **backwash** largely causes deposition on lowland coasts.
- (b) The geology of the coastline. Resistant rocks along the coasts tend to form coastal uplands projecting into the sea known **headlands**. Soft or weak rocks are eroded and forms **bays**. In Africa, they are common in areas where mountains reach the ocean for instance the Cape Ranges in South Africa and Atlas Mountains in northwest Africa.
- (c) Eustatic changes in seawater. These are relative movements in sea level. Rise and fall in sea level influence both sea erosion and deposition.
- (d) Climate – past or present climates along the coast has influenced coastal landforms. Warm climate in the Tropics has lead to coral and mangrove coasts. Fjords have resulted from ice age cold climates.
- (e) Man's influence – man's activities such as building canals, artificial harbours and dredging estuaries leads to changes in the original coastal configuration.

9.5 Landforms of Wave Erosion

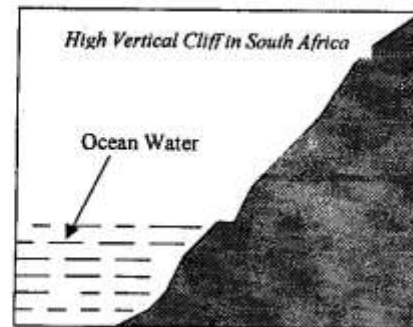
Wave erosion is pronounced on highland or raised coasts. One method of wave erosion is by **hydraulic action** that involves the force of water, which hits the coastline and compresses air in the cracks or joints during the swash. When the waves retreats, sometimes explosively during the backwash, this leads to expansion of air. The compression and expansion causes erosion. **Corrasion** takes place when waves use pebble, boulders and sand as tools of coastal erosion. Lastly, **attrition** involving breaking up of pebbles, boulders and sand, as they hit the coastline and among themselves may also occur.

Features formed by wave erosion include but are not limited to:

(a) Cliffs

This is a rock face along the coast where land rises steeply inland. They normally occur at the front or sides of the headlands. Headlands are part of an irregular coastline where land protrudes out into the sea.

Figure 24: A High Vertical Cliff



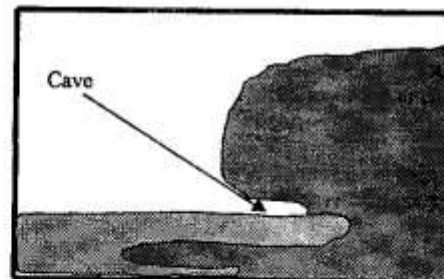
A cliff develops as the base of the coastal steep rock surface experiences undercutting during high tide. The rock face steepens as its base is attacked by wave action while the area above water is worn out by weathering. Materials that fall off become tools for further erosion. On Mombasa Island and Cape point in Kenya and South Africa respectively can be observed a number of cliffs.

(b) Caves

A cave is a tunnel-like opening at the base of a cliff. It is formed where there are weaknesses in the rock such as faults, joints, bedding planes or belts of less resistant rocks.

Wave action constantly attacks the lines of weakness and opens or widens them. Waves put a lot of pressure on the cliff rocks causing air in the cracks to compress and expand as the water hits and retreats. The alternate compression and expansion enlarges the lines of weakness into holes and later caves.

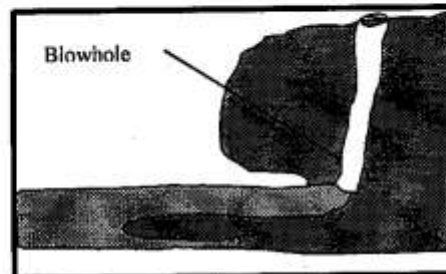
Figure 25: A Cave



(c) Blowholes

If a line of weakness extends from the base of a cliff to the surface and it is widened by wave erosion, a blowhole forms. It is also formed when a cave is further eroded by alternate compression and expansion until this process reaches the top of a cliff. The coast of Reunion, Indian Ocean has a number of blowholes.

Figure 26: A Blow Hole



(d) A Geo – This is formed when the roof of a cave collapses. With this collapse, a thin inlet, known as a geo is formed.

(e) An Arch

This feature is formed on headland coasts, where a mass of resistant rock protrudes into the sea. Erosion attacks the headland from both sides. If the waves erode right through the headland, a natural bridge is formed called an arch.

(f) A stack

If the top of an arch collapses, part of the headland is cut off and forms a stack, a mass of rock cut from the main land.

It is important to note that erosional features are not found on all highland or cliff coasts. They tend to be common on stormy wave coasts.

9.6 Depositional Features

Depositional features are most common on lowland coasts. In this areas, wave deposition far outweighs erosion. Some of the depositional features include but are not limited to:

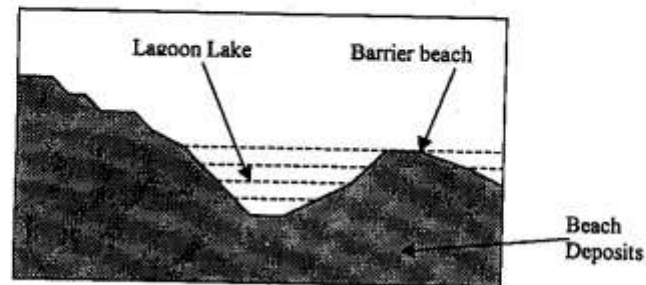
(a) Beaches

A beach is an accumulation of sand formed where the coast is low-lying especially in bays. Beach boundaries extend between the low water mark and the maximum level mark of the high waves.

Beach construction is most active during calm weather. During storms, sand deposits are swept away by waves destroying the beaches. Nyali beach in Mombasa is an example of these kinds of sand deposits.

In some places, some types of beaches such as barrier beach, part of the ocean water is enclosed forming a lagoon lake.

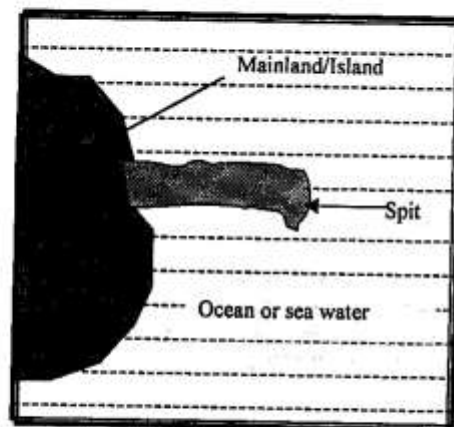
Figure 27: A Lagoon Lake behind a Barrier Beach



(b) Spit

A spit is a low, narrow ridge of sand or pebbles. It is joined to the mainland or island on one side while the other end terminates to the sea. It is also formed by wave deposition. The tip of a spit may be bent depending on longshore drift. The mouth of river Tana in Kenya has a growing spit. Other examples include Walvis Bay in South Africa. And Mdjerda spit in Tunisia.

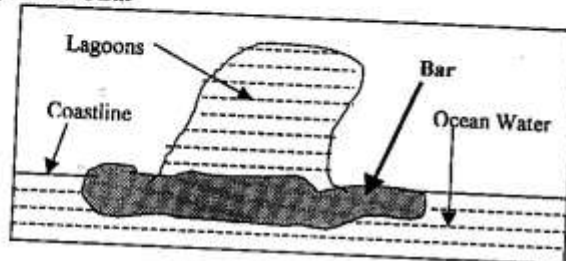
Figure 28: A Spit



(c) Bar

This is a ridge of material deposits lying parallel or almost parallel to the coast. It is not attached to the mainland as a spit. Formed by wave deposition. It may occur across a bay forming a shallow lake called a lagoon. A marsh of plants may occupy the shallow lagoon. Lagos in West Africa, Nigeria lies on an Island found on a lagoon. The Nile delta also has a number of lagoons.

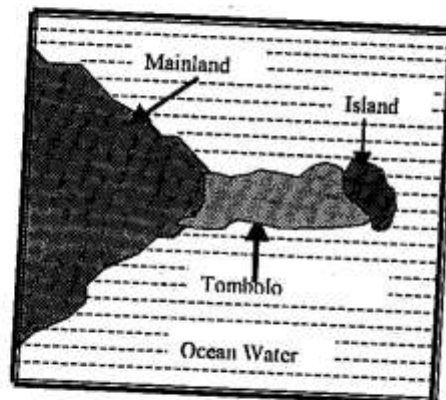
Figure 29: A Bar



(d) Tombolo

This is formed when a bar joins an island to the mainland due to deposition of ocean materials or deposits. A large Tombolo occurs in North East Somalia and extends up to a length of 20 km. Dakar in Senegal is located on a large Tombolo. The same applies for Free Town in Sierra Leone.

Figure 30: A Tombolo



9.7 Types of Coasts

(a) Ria Coast

A ria is a feature formed where the sea or ocean reaches far inland and the river mouths are very deep. It is formed when sea level rises and invades low parts of the land along a river valley.

It is also formed if parts of the land sink or submerge. In both cases, the sea invades the land and finds a new level. Freetown and Banjul both lie at the mouths of rias. Mombasa lies on an island in the middle of a ria.

(b) Coral Coasts

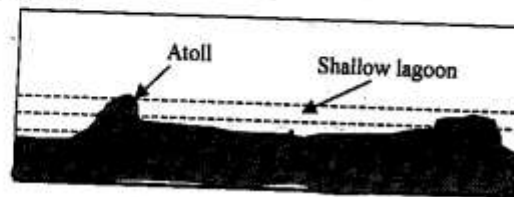
These are coasts mainly composed of organic material. Coral is a limestone rock formed from the skeletons of very small marine organisms called coral polyps that thrive under the following conditions:

- Warm water temperatures (21°C)
- Sediment free salt water
- Clear water where sunlight is able to penetrate up to 50 m deep.
- Presence of plentiful supply of planktons.

A coral mass on a coral coast is known as a reef. There are three types of reefs. They include a **fringing a reef**, **barrier reef**, and **atoll**. A fringing reef is a coral mass that is connected to the coastline. It may enclose a shallow lagoon. A barrier reef is located several kilometers from the shore. It has a deep lagoon.

An atoll is a circular coral reef enclosing a shallow lagoon.

Figure 31: An Atoll



(c) Headland Coasts

Headlands are formed when parts of a rocky coastline is characterized by rocks resistant to erosion. The coast has rocks of differing strengths. Due to wave erosion, resistant rocks form headlands while less resistant rocks form bays.

Figure 32: A Headland Coast

